بسم الله الرحمن الرحيم

SET THEORY

- **A *set* is an unordered collection of objects.**
  - **the students in this class**
  - **the chairs in this room**

- **The objects in a set are called the elements, or members of the set. A set is said to contain its elements.**
- **The notation $a \in A$ denotes that *a* is an element of the set *A*.**
- **If *a* is not a member of *A*, write $a \notin A$**

Set Theory
Basics of Sets

What is a set?

A set is a well-defined collection of distinct objects.

Object: An object could be anything. It can be something we can touch or see or it can be an idea or a concept.

Examples: A set of all facts learned in discrete mathematics course.
A collection of pens.
A collection of cars.
A set of odd numbers divisible by 2.
A set of vowels of English alphabet.

But, what is not a set?

"A collection of beautiful songs."

What? But why?

This is where the term well-defined came into picture.

Well-defined set: A set is considered to be well-defined if it is possible to establish that any given object belongs to the set.

In our example, "A set of beautiful songs"
"beautiful songs" is not well-defined. The definition of beautiful song changes from person to person.

Therefore, a set of beautiful songs is not well-defined and hence it is not a set.

More examples:

1. A collection of great people of the world.
2. A set of beautiful flowers.
3. A collection of best football players in the world.
4. A collection of most dangerous animals found in the forest.
5. A collection of the most talented boys in your class.

SOME OTHER EXAMPLES

- **State whether each of the following collections is a well defined set. Give a reason for you answer.**

1. **{A book well-liked by my classmates}**

Answer: No, b/c a book may be well-liked by some, but not others.

2. **{A naughty pupils in my class}**

Answer: No, b/c some teachers and pupils may consider a certain pupil naughty , while other may not

3. **{A mathematics Teacher in my class}**

Answer: Yes, b/c it is clear whether someone is a Mathematics teacher in the class

SOME OTHER EXAMPLES

- **State whether each of the following collections is a well defined set. Give a reason for you answer.**

1. **{A movie well liked by my classmate}……….NO**

2. **{A 14 year old pupil in my class}……………...YES**

3. **{A English Teacher in my class}……………….YES**

Distinct objects:

We can have a set with duplicate objects.
But a set with duplicate objects is similar to a set with distinct objects.

For example: $A = \{1, 2, 2, 3, 3, 3\}$
$B = \{1, 2, 3\}$

$A = B$

Eventually, we end up with a set without duplicate elements.
This is the reason why the definition

"A set is a well-defined collection of distinct objects" holds true.

Set Membership

Any object belonging to a set is called a <u>member</u> or an <u>element</u> of that set.

We will represent sets by uppercase letters and lowercase letters will be used to represent the elements of the set.

If a is an element of set A then

$a \in A$ or a is in A

"belongs to" symbol.

If there exist an element b that does not belong to set A, then we express this fact by

$b \notin A$ or b is not in A

"does not belong to" symbol

Set Representation

Three ways to represent a set:

1. List representation.
2. Predicate representation.
3. Missing element representation.

1. List representation:

Let us suppose we have a set A with elements 1, 2, 3, a and b.
Generally, a set is represented by listing all the elements of it. Here, set A is represented by

$$A = \{1, 2, 3, a, b\}$$

Here elements are simply listed within the pair of brackets ({})

REPRESENTATION OF SET

TABULAR FORM

We list all the elements of a set, separated by commas and enclosed within braces or curly brackets{}.

EXAMPLES

In the following examples we write the sets in Tabular Form.

$A = \{1, 2, 3, 4, 5\}$ is the set of first five **Natural Numbers**.
$B = \{2, 4, 6, 8, \ldots, 50\}$ is the set of **Even numbers** up to 50.
$C = \{1, 3, 5, 7, 9, \ldots\}$ is the set of **positive odd numbers.**

DESCRIPTIVE FORM:

We state the elements of a set in words.

EXAMPLES

Now we will write the above examples in the Descriptive Form.

A = set of first five Natural Numbers. (Descriptive Form)
B = set of positive even integers less or equal to fifty. (Descriptive Form)
C = set of positive odd integers. (Descriptive Form)

SET BUILDER FORM:

We write the common characteristics in symbolic form, shared by all the elements of the set.

EXAMPLES:

Now we will write the same examples which we write in Tabular as well as Descriptive Form ,in Set Builder Form .

$A = \{x \in N \mid x \leq 5\}$ (Set Builder Form)
$B = \{x \in E \mid 0 < x \leq 50\}$ (Set Builder Form)
$C = \{x \in O \mid 0 < x\}$ (Set Builder Form)

1. Roster or Tabular Method

- **Definition:** All the elements of the set are listed, separated by commas, and enclosed in curly braces `{}`.
- **Example:**
  1. $A = \{1, 2, 3, 4, 5\}$ (Set of first 5 natural numbers)
  2. $B = \{\text{apple, banana, cherry}\}$ (Set of fruits)
  3. $C = \{x \mid x \in \mathbb{N}, x < 10\} = \{1, 2, 3, \ldots, 9\}$

2. Set-Builder Method

- **Definition:** The elements of the set are described using a property or rule instead of listing them.
- **Example:**
  1. $A = \{x \mid x \text{ is an even number less than } 10\}$
  2. $B = \{x \mid x \text{ is a vowel in the English alphabet}\}$
  3. $C = \{x \mid x \in \mathbb{Z}, -3 \leq x \leq 3\}$

3. Descriptive Method

- **Definition:** The set is described in words rather than listing elements or using a rule.
- **Example:**
  1. A: "The set of first 5 odd numbers."
  2. B: "The set of all planets in the solar system."
  3. C: "The set of all integers divisible by 5."

Inclusion and Equality

Inclusion:

Let A and B are two sets. If every element of A is an element of B, then A is called a subset of B or A is said to be included in B.

$$A \subseteq B \quad \text{(A is a subset of B)}$$

or

$$B \supseteq A \quad \text{(B is a superset of A)}$$

Example: $A = \{1, 2, 3\}$ $\quad$ $A \subseteq B$ but $B \supseteq A$

$B = \{1, 2, 3, 4, 5\}$ $\quad$ (Note: $B \not\subset A$)

Note: $A \subseteq B$ if and only if the quantification $\forall x(x \in A \rightarrow x \in B)$ is true.

Why?

Example: $A = \{1, 2\}$ $\quad$ Consider all elements of A

$B = \{1, 3, 5\}$

$1 \in A$ and $1 \in B$

$2 \in A$ but $2 \notin B$

$\therefore x \in A \rightarrow x \in B$ is not true for all elements of A.

Important properties of set inclusion:

1. Reflexivity: $A \subseteq A$

   Example: $A = \{1, 2, 3\}$

   It is true that A is itself the subset of A.

2. Transitivity: $(A \subseteq B) \land (B \subseteq C) \rightarrow (A \subseteq C)$

   Example: $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 5\}$, and $C = \{1, 2, 3, 5, 7\}$

   it is clear that $A \subseteq B$ and $B \subseteq C$

   Also, it is clear that $A \subseteq C$

$\therefore$ set inclusion is both reflexive and transitive.

Inclusion Vs Membership

Lets try to understand the difference between inclusion and membership with the help of an example.

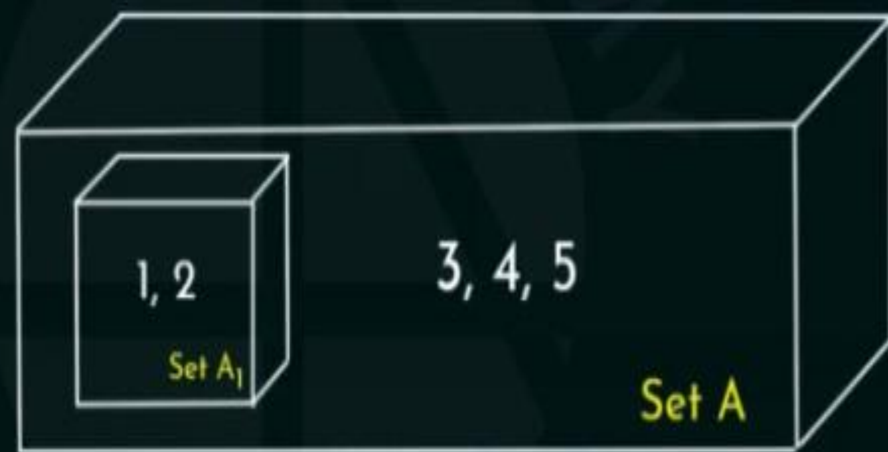
Example: Let $A = \{\{1, 2\}, 3, 4, 5\}$ and $B = \{1, 2, 3, 4, 5\}$

Which of the following is true?

$1 \in B$? true.

$1 \in A$? false.

Let us assume that $\{1, 2\}$ is represented by the name "Set A_1"



Try to understand this analogy.

We have a box named "Set A." After opening the box, we can see 4 different objects. One is a box named "Set A_1" and the rest are the elements 3, 4 and 5.

So, opening the box is associated with knowing the members of the set.

Here, we have opened the box named "Set A". After opening the box, we can only see the box "Set A_1" and not its elements. Hence, we don't know what is there inside set A_1

So, is it true that $1 \in A$?

No. The elements of Set A are Set A_1, 3, 4, 5.

But $1 \in A_1$

More questions:

$A = \{\{1, 2\}, 3, 4, 5\}$ and $B = \{1, 2, 3, 4, 5\}$

>> $\{1, 2\} \in A$?

True. $\{1, 2\}$ is a set within set A. Therefore, $\{1, 2\} \in A$.

>> $\{3, 4\} \subseteq A$?

True. Whenever it is required to answer if a particular set is a subset of a different set, see the elements of the given set and compare it with the elements of the other set.

Here, the given set is $\{3, 4\}$.

Ask this: $3 \in A$? Yes
$4 \in A$? Yes

$\therefore \{3, 4\} \subseteq A$

>> $\{1, 6\} \subseteq B$?

False. Ask yourself: $1 \in B$? Yes.
$6 \in B$? No.

$\therefore \{1, 6\} \not\subseteq B$

$A = \{\{1, 2\}, 3, 4, 5\}$ and $B = \{1, 2, 3, 4, 5\}$

>> $1 \subseteq B$?

False. 1 is not a set itself.

>> $1 \in B$?

True. 1 is the element in B

$\therefore 1 \in B$

>> $\{1, 2\} \subseteq A$?

False. Ask yourself: $1 \in A$? No.

(Note: 1 belongs to set $\{1, 2\}$ contained within set A. It does not belong to A)

$2 \in A$? No.

$\therefore \{1, 2\} \not\subseteq A$

>> $\{\{1, 2\}\} \subseteq A$?

True. Ask yourself: $\{1, 2\} \in A$? Yes.

$\therefore \{\{1, 2\}\}$ is the subset of A.

>> $\{\{1, 2\}, 3, 4, 5\} \subseteq A$?

True. In fact, the given set is equal to A.

Inclusion Vs Membership

(Solved Problem)

Given $S = \{2, a, \{3\}, 4\}$ and $R = \{\{a\}, 3, 4, 1\}$. Indicate whether the following are true or false.

a) $\{a\} \in S$

False.

b) $\{a\} \in R$

Yes. Set $\{a\}$ is member of set R.

c) $\{a, 4, \{3\}\} \subseteq S$

Ask yourself: $a \in S$? Yes.
$4 \in S$? Yes.
$\{3\} \in S$? Yes.

Therefore, $\{a, 4, \{3\}\}$ is the subset of set S.

d) $\{\{a\}, 1, 3, 4\} \subset R$

No. because $\{\{a\}, 1, 3, 4\} = R$ and is not a proper subset of R.

e) $R = S$

No. $2 \in S$ and $2 \notin R$, $a \in S$ and $a \notin R$, $\{3\} \in S$ and $\{3\} \notin R$ & $4 \in S$ and $4 \in R$

f) $\{a\} \subseteq S$

Ask yourself: $a \in S$? Yes.

g) $\{a\} \subseteq R$

Ask yourself: $a \in R$? No.

SETS OF NUMBERS:

1. **Set of Natural Numbers**

   N = {1, 2, 3, … }

2. **Set of Whole Numbers**

   W = {0, 1, 2, 3, … }

3. **Set of Integers**

   Z = {…, -3, -2, -1, 0, +1, +2, +3, …}

   = {0, ±1, ±2, ±3, …}

{"Z" stands for the first letter of the German word for integer: Zahlen.}

4. **Set of Even Integers**

   E = {0, ± 2, ± 4, ± 6, …}

5. **Set of Odd Integers**

   O = {± 1, ± 3, ± 5, …}

6. **Set of Prime Numbers**

   P = {2, 3, 5, 7, 11, 13, 17, 19, …}

7. **Set of Rational Numbers (or Quotient of Integers)**

   $$Q = \{x \mid x = \frac{p}{q},\ p, q \in Z,\ q \neq 0\}$$

8. **Set of Irrational Numbers**

   $\overline{Q} = Q' = \{\, x \mid x \text{ is not rational}\}$

   For example, $\sqrt{2}$, $\sqrt{3}$, π, e, etc.

9. **Set of Real Numbers**

   $R = Q \cup Q'$

10. **Set of Complex Numbers**

   C = {z | z = x + *i*y; x, y ∈ R} Here, $i = \sqrt{-2}$

FINITE AND INFINITE SETS:

A set **S** is said to be **finite** if it contains exactly m distinct elements where m denotes some non negative integer.

In such case we write $|S| = m$ or $n(S) = m$

A set is said to be **infinite** if it is not finite.

EXAMPLES:

1. The set S of letters of English alphabets is finite and $|S| = 26$
2. The null set $\varnothing$ has no elements, is finite and $|\varnothing| = 0$
3. The set of positive integers $\{1, 2, 3, \ldots\}$ is infinite.

EXERCISE:

Determine which of the following sets are finite/infinite.

1.	A = {month in the year}	**FINITE**
2.	B = {even integers}	**INFINITE**
3.	C = {positive integers less than 1}	**FINITE**
4.	D = {animals living on the earth}	**FINITE**
5.	E = {lines parallel to x-axis}	**INFINITE**
6.	$F = \{x \in \mathbf{R} \mid x^{100} + 29x^{50} - 1 = 0\}$	**FINITE**
7.	G = {circles through origin}	**INFINITE**

Universal Set, Null Set, and Singleton Set

Universal Set:

A universal set is a set which includes every set under consideration.
A universal set is represented by E.
For any predicate, P(x)

$$E = \{x \mid P(x) \vee \neg P(x)\}$$

The universal set is same as universe of discourse.

Null Set:

A set which does not contain any element is called a null set or empty set.
It is denoted by ϕ or $\{\}$.

$$\phi = \{x \mid P(x) \wedge \neg P(x)\}$$

For example: A set of all positive integers which are both even and odd.

Singleton Set:

A singleton set is a set with exactly one element.

For example: $A = \{2\}$
$B = \{\phi\}$

please note that $\{\phi\} \neq \phi$

$\{\phi\}$ consists of one element which is the null set itself while there is no element inside ϕ.

Null Set

(Solved Problem)

Determine whether the following statements are true or false.

a) $\phi \in \{\phi\}$

True. ϕ is an empty set and is also a member of set $\{\phi\}$.

b) $\phi \in \{\phi, \{\phi\}\}$

True.

c) $\{\phi\} \in \{\phi\}$

False. $\{\phi\}$ is not a member of the $\{\phi\}$ because there is only one element in the set $\{\phi\}$ which is ϕ not $\{\phi\}$.

d) $\{\phi\} \in \{\{\phi\}\}$

True. $\{\phi\}$ is the member of the set $\{\{\phi\}\}$.

e) $\{\phi\} \subset \{\phi, \{\phi\}\}$

True. Ask yourself: $\phi \in \{\phi, \{\phi\}\}$? Yes. ϕ is an element of the set $\{\phi, \{\phi\}\}$

f) $\{\{\phi\}\} \subset \{\phi, \{\phi\}\}$

True. Ask yourself: $\{\phi\} \in \{\phi, \{\phi\}\}$? Yes. $\{\phi\}$ is an element of the set $\{\phi, \{\phi\}\}$

g) $\{\{\phi\}\} \subset \{\{\phi\}, \{\phi\}\}$

False. The above statement is equivalent to $\{\{\phi\}\} \subset \{\{\phi\}\}$.

Important Theorem on Non-Empty Set

Theorem: Every non-empty set A is guaranteed to have at least two subsets, the empty set and the set A itself i.e. $\phi \subseteq A$ and $A \subseteq A$.

Proof: (i) $\phi \subseteq A$

If ϕ is a subset of A, then from the definition of subset:

$\forall x(x \in \phi \rightarrow x \in A)$

$x \in \phi$ is always false because no element can be a member of an empty set.

Therefore, the implication $x \in \phi \rightarrow x \in A$ will always be true.

$\boxed{\text{Hence, } \phi \subseteq A.}$

(ii) $A \subseteq A$

If A is a subset of A, then from the definition of subset:

$\forall x(x \in A \rightarrow x \in A)$

No matter what x we choose, if $x \in A$ is true, $x \in A$ must be true.

Therefore, the implication $x \in A \rightarrow x \in A$ will always be true for every element x.

$\boxed{\text{Hence, } A \subseteq A}$

Power Set

For a particular set A, a collection of all subsets of set A is called the power set of set A.
Usually, a power set of set A is denoted by $P(A)$ or 2^A.

$$P(A) = \{x \mid x \subseteq A\}$$

Examples: $P(\phi) = \{\phi\}$

$A = \{a\}, \quad P(A) = \{\phi, \{a\}\}$

$B = \{a, b\}, \quad P(B) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$

$C = \{a, b, c\}, \quad P(C) = \{\phi, \{a\}, \{b\}, \{c\}, \{a, b\}, \{b, c\}, \{a, c\}, \{a, b, c\}\}$

This is the reason why power set of A is sometimes represented by 2^A.

In set A, we have only 1 element.

Therefore, $P(A)$ has $2^1 = 2$ elements.

In set B, we have 2 elements.

Therefore, $P(B)$ has $2^2 = 4$ elements.

Therefore, if we have n elements in set A, then $P(A)$ must have 2^n elements.

Power Set

(Solved Problems)

Problem 1: Give the power sets of the following

a) $\{a, \{b\}\}$
b) $\{1, \phi\}$
c) $\{X, Y, Z\}$

Solution: a) Let $A = \{a, \{b\}\}$

$P(A) = \{\phi, \{a\}, \{\{b\}\}, \{a, \{b\}\}\}$

b) Let $B = \{1, \phi\}$

$P(B) = \{\phi, \{1\}, \{\phi\}, \{1, \phi\}\}$

c) Let $C = \{X, Y, Z\}$

$P(C) = \{\phi, \{X\}, \{Y\}, \{Z\}, \{X, Y\}, \{Y, Z\}, \{X, Z\}, \{X, Y, Z\}\}$

Power Set

(Solved Problems)

Problem 1: Give the power sets of the following

a) $\{a, \{b\}\}$
b) $\{1, \phi\}$
c) $\{X, Y, Z\}$

Solution: a) Let $A = \{a, \{b\}\}$

$P(A) = \{\phi, \{a\}, \{\{b\}\}, \{a, \{b\}\}\}$

b) Let $B = \{1, \phi\}$

$P(B) = \{\phi, \{1\}, \{\phi\}, \{1, \phi\}\}$

c) Let $C = \{X, Y, Z\}$

$P(C) = \{\phi, \{X\}, \{Y\}, \{Z\}, \{X, Y\}, \{Y, Z\}, \{X, Z\}, \{X, Y, Z\}\}$

Problem 2: Find the power set of each of these sets, where a and b are distinct elements.

a) $\{a\}$ b) $\{a, b\}$ c) $\{\phi, \{\phi\}\}$

Solution: a) Let $A = \{a\}$, $P(A) = \{\phi, \{a\}\}$

b) Let $B = \{a, b\}$, $P(B) = \{\phi, \{a\}, \{b\}, \{a, b\}\}$

c) Let $C = \{\phi, \{\phi\}\}$, $P(C) = \{\phi, \{\phi\}, \{\{\phi\}\}, \{\phi, \{\phi\}\}\}$

The Intersection Operator

- For sets A, B, their *intersection* $A \cap B$ is the set containing all elements that are simultaneously in A **and** ("$\wedge$") in B.
- Formally, $\forall A,B$: $A \cap B \equiv \{x \mid x \in A \wedge x \in B\}$.
- Note that $A \cap B$ is a subset of A **and** it is a subset of B:
  $\forall A, B$: $(A \cap B \subseteq A) \wedge (A \cap B \subseteq B)$

Problem 3: How many elements does each of these sets have where a and b are distinct elements?

a) $P(\{a, b, \{a, b\}\})$

b) $P(\{\phi, a, \{a\}, \{\{a\}\}\})$

c) $P(P(\phi))$

Solution:

a) The set $\{a, b, \{a, b\}\}$ has 3 elements.
$P(\{a, b, \{a, b\}\})$ contains all possible subsets of $\{a, b, \{a, b\}\}$.
Therefore, $P(\{a, b, \{a, b\}\})$ has $2^3 = 8$ elements.

b) Let $A = \{\phi, a, \{a\}, \{\{a\}\}\}$ has 4 elements.
Therefore, $P(A)$ has $2^4 = 16$ elements.

c) The set ϕ has no elements.
$P(\phi)$ has 1 element which is set ϕ.
Therefore, $P(P(\phi))$ must have $2^1 = 2$ elements.

$P(P(\phi)) = P(\{\phi\}) = \{\phi, \{\phi\}\}$

Cardinality of Sets

Cardinality of a set means size of a set.
Cardinality of set A is denoted by $|A|$.

Example: A set of all odd positive integers less than 10.

$A = \{1, 3, 5, 7, 9\}$ $\boxed{|A| = 5}$

Example 2: Cardinality of null set is 0. $|\Phi| = 0$

[GATE Problem] [GATE 2015]

The cardinality of the power set of $\{0, 1, 2, \ldots, 10\}$ is?

Solution: Let say $A = \{0, 1, 2, \ldots, 10\}$ Total 11 elements in set A

Therefore, $|P(A)| = 2^{11} = 2048$.

Cartesian Product

Sometimes, we want a collection of ordered pairs.
Let's say we are interested in finding all possible combinations of X and Y coordinates where X and Y values should not exceed 3.

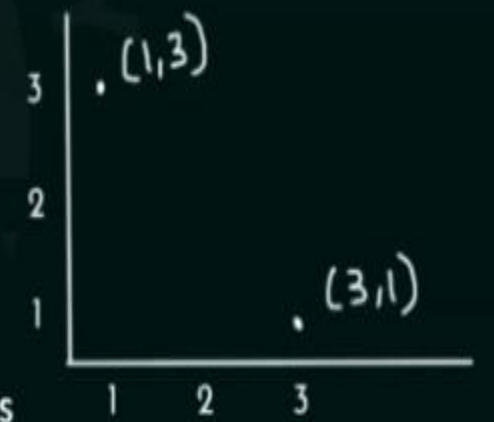
$\{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$

Here, (a, b) for any a and b is called an ordered pair.

Note: (a, b) and (b, a) are not equal.

Therefore, in an ordered pair, order does matter.

Example:



Definition of Cartesian Product:

Let A and B are two sets. The cartesian product of A and B, denoted by AXB, is the set of all ordered pairs (a, b), where $a \in A$ and $b \in B$.

$$AXB = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

Example 1: What is the cartesian product of $A = \{1, 2\}$ and $B = \{a, b, c\}$?

Solution: $A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$

Note 1: $AXB \neq BXA$

Example: $A = \{1, 2\}$ and $B = \{3, 4\}$

$A \times B = \{(1,3), (1,4), (2,3), (2,4)\}$

$B \times A = \{(3,1), (3,2), (4,1), (4,2)\}$

Note 2: For any two non-empty sets A and B, if there are n elements in A and m elements in B, then the number of ordered pairs in AXB will be <u>nm</u>.

Example: $A = \{a, b\}$ and $B = \{p, q, r\}$

$n(\underline{AXB}) = 2 \times 3 = 6$

$(\underline{2}\ \ \underline{3}) \quad 2 \times 3 = 6$

Note 3: $\phi \times A = A \times \phi = \phi$

From the definition,

$\phi \times A = \{(a, b) \mid a \in \phi \text{ and } b \in A\} = \phi$

There is no such pair (a, b) where $a \in \phi$ and $b \in A$.

Similarly, $A \times \phi = \phi$

Therefore, $\phi \times A = A \times \phi = \phi$

Cartesian Product (Solved Problems)

Question 1: What is the cartesian product AXB, where A is the set of all courses offered by the computer science department at a university (i.e. Data Structures, Algorithms) and B is the set of computer science professors at this university (i.e. Mark, Allen)?

Solution: A = {DS, Algo}   B = {Mark, Allen}

AXB = {(DS, Mark), (DS, Allen), (Algo, Mark), (Algo, Allen)}

Question 2: What is the cartesian product AXBXC, where A is the set of all airlines (i.e. Indigo, Air Asia) and B and C are both the set of all cities in India (i.e. Mumbai, Hyderabad)?

Solution: A = {Ind, AA}
B = {Mum, Hyd}
C = {Mum, Hyd}

AXBXC = {(Ind, Mum, Mum), (Ind, Mum, Hyd), (Ind, Hyd, Mum), (Ind, Hyd, Hyd), (AA, Mum, Mum), (AA, Mum, Hyd), (AA, Hyd, Mum), (AA, Hyd, Hyd)}

Question 3: Show by means of an example that $A \times B \times C \neq (A \times B) \times C$

Solution:

Let $A = \{1, 2\}$, $B = \{3, 4\}$, $C = \{5, 6\}$

$A \times B \times C = \{(a, b, c) \mid a \in A \wedge b \in B \wedge c \in C\}$

$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$

$(A \times B) \times C = \{(x, c) \mid x \in A \times B \wedge c \in C\}$

$(A \times B) \times C = \{((a, b), c) \mid (a \in A \wedge b \in B) \wedge c \in C\}$

$A \times B \times C = \{(1, 3, 5), (1, 3, 6), (1, 4, 5), (1, 4, 6), (2, 3, 5), (2, 3, 6), (2, 4, 5), (2, 4, 6)\}$

$(A \times B) \times C = \{((1, 3), 5), ((1, 3), 6), ((1, 4), 5), ((1, 4), 6), ((2, 3), 5), ((2, 3), 6),$
$((2, 4), 5), ((2, 4), 6)\}$

Set Operations

(Intersection and Union with Venn Diagrams)

Definition of Intersection:

The intersection of any two sets A and B, denoted by $A \cap B$, is the set consisting of all the elements which belong to both A and B.

$$A \cap B = \{x \mid (x \in A) \wedge (x \in B)\}$$

Representing Intersection of Two Sets Pictorially using Venn Diagrams

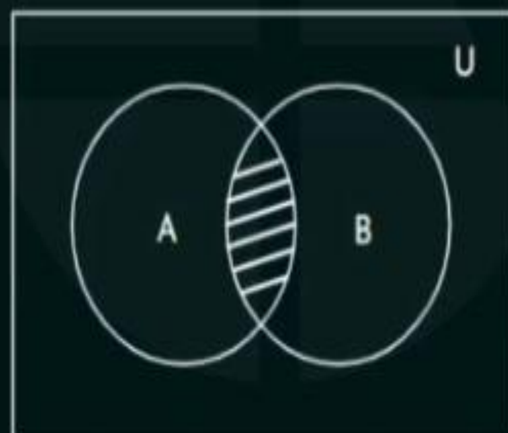
A Venn diagram is a diagram used to illustrate the logical relationship between two or more sets by using overlapping circles or other shapes.

Basic Overview:

A set is usually represented by a circle and the elements of the set lies within the circle.

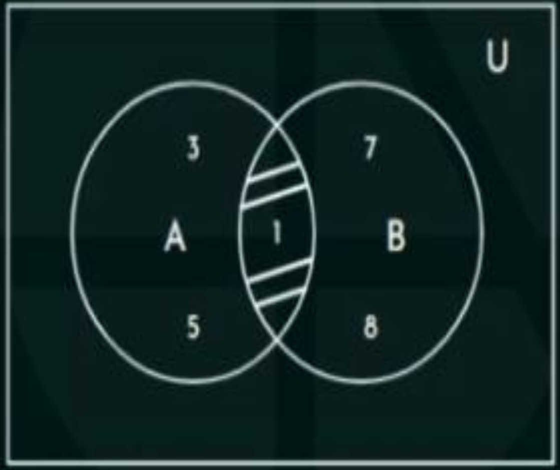
A universal set or universe of discourse is represented by a rectangle. Every element under consideration lies within the rectangle.

Pictorial Representation of Intersection of sets A and B



Shaded region is the common area between A and B i.e. $A \cap B$.

Example: Let $A = \{1, 3, 5\}$ and $B = \{1, 7, 8\}$

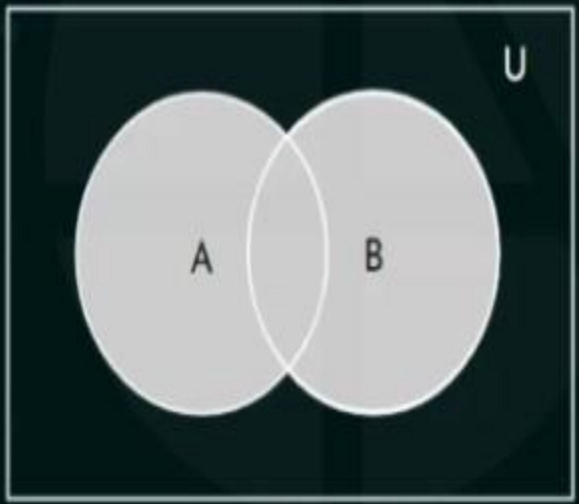


Definition of Union

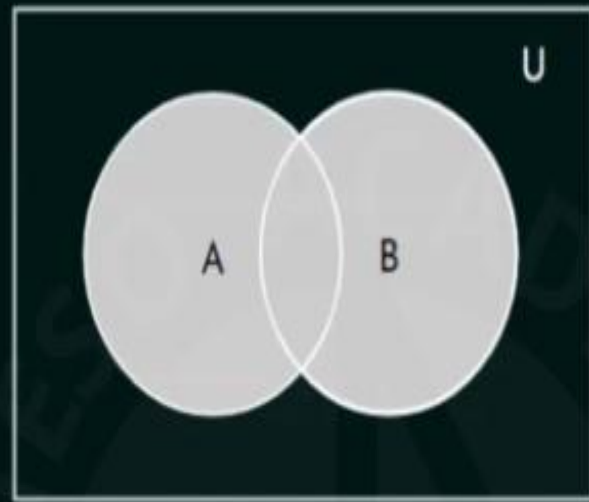
For any two sets A and B, the union of A and B, denoted by $A \cup B$, is the set of all elements which are members of the set A or set B or both.

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

Venn Diagram of Union of Sets

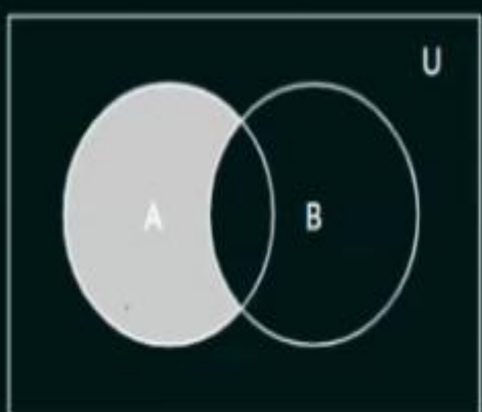


Shaded region is representing union of A and B.

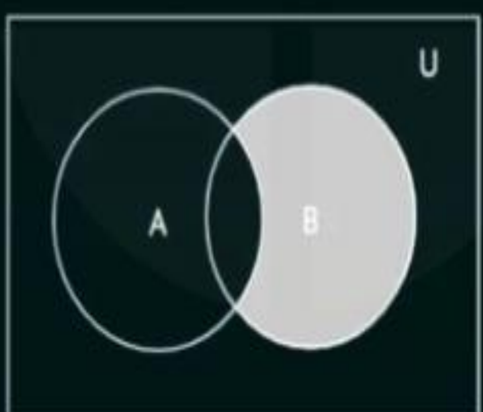


Shaded region is representing union of A and B.

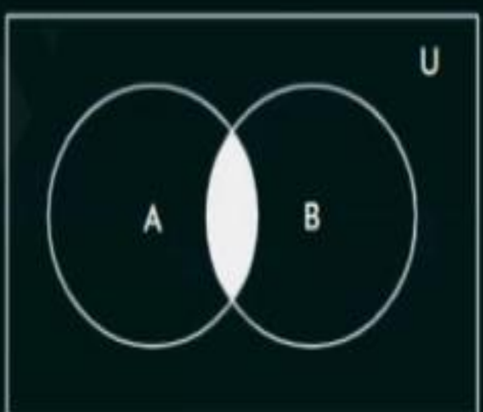
Either Circle A



or Circle B



or Both



Example: $A = \{1, 3, 5\}$
$B = \{1, 7, 8\}$

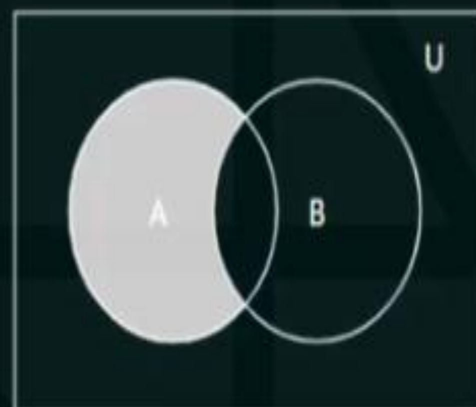
$A \cup B = \{1, 3, 5, 7, 8\}$

Set Difference and Complement

For any two sets A and B, the difference of A and B, denoted by A - B, is the set containing those elements that are in set A but not in set B.

$$A - B = \{x \mid x \in A \wedge x \notin B\}$$

Venn Diagram Representation of A - B



Shaded region represents A but not B

The difference of A and B is also called the complement of B with respect to A.

Example 1: Let say, $A = \{1, 3, 5\}$ and $B = \{1, 2, 3\}$

$A - B = \{5\}$ (A but not B)

Example 2: Let say A represents a set of all computer science students at a University and B represents a set of all students who are studying discrete mathematics.

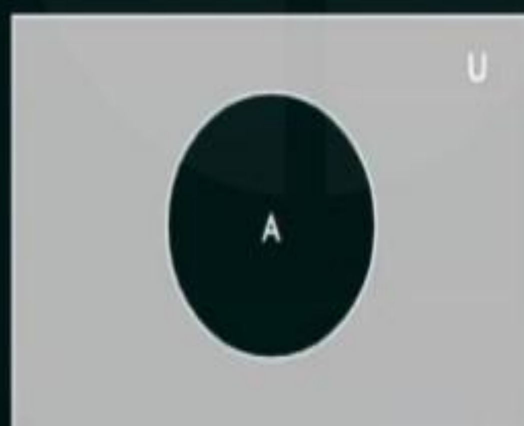
then, A - B will represent a set of all computer science students who are not studying discrete mathematics.

Complement of a Set

Let U be the Universal set.
The complement of set A, denoted by A', is the complement of A with respect to U.
In other words, complement of set A is U - A.

$$A' = \{x \mid x \notin A\}$$

Venn Diagram for the Complement of Set A



Shaded region represents A'

Set Operations

(Solved Problems)

Question 1: Let A be the set of students who live within one mile of school and B be the set of students who walk to classes. Describe the students in each of these sets.

a) $A \cup B$ b) $A \cap B$ c) $A - B$ d) $B - A$

Solution: Let say $A = \{Mark, Allen, Roy\}$
$B = \{Allen, John, Mark, Henry\}$

a) $A \cup B$ contains all students who live within one mile of school or who walk to classes.
i.e. $A \cup B = \{Mark, Allen, Roy, John, Henry\}$

b) $A \cap B$ contains all students who live within one mile of school and who walk to classes i.e. $A \cap B = \{Mark, Allen\}$

c) $A - B$ contains all students who live within one mile of school, but who do not walk to classes i.e. $A - B = \{Roy\}$

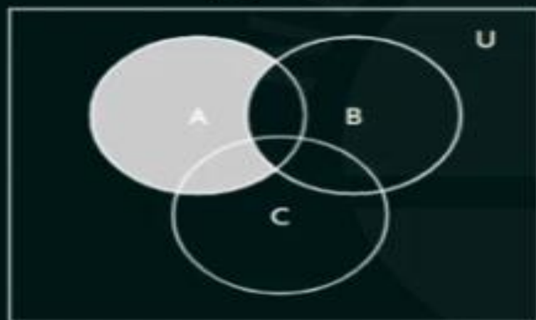
d) $B - A$ contains all students who walk to classes, but do not live within one mile of school i.e. $B - A = \{John, Henry\}$

Question 2: Let A, B and C be non-empty sets and let $X = (A - B) - C$ and $Y = (A - C) - (B - C)$. Which one of the following is True?

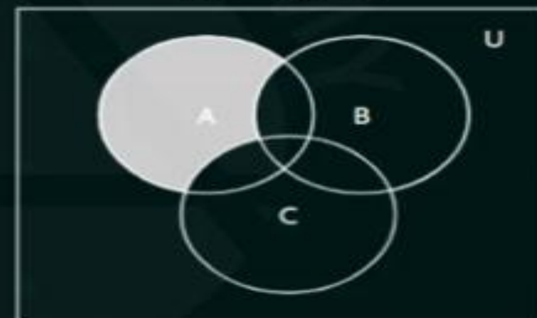
(A) $X = Y$ (B) $X \subset Y$ (C) $Y \subset X$ (D)None of the mentioned

Solution:

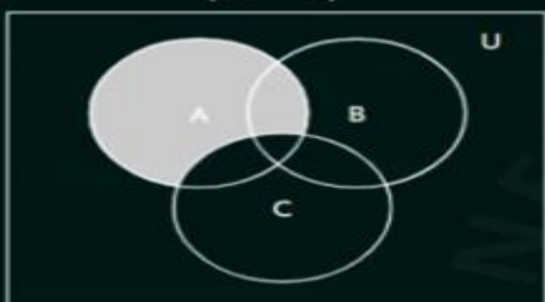
(A - B)



(A - B) - C

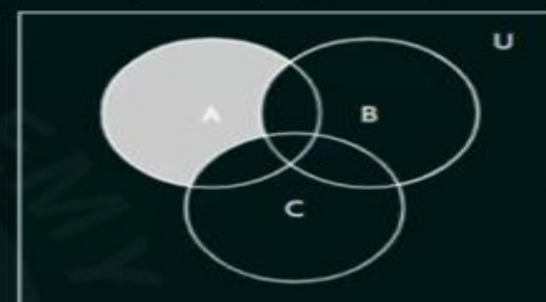


(A - C)

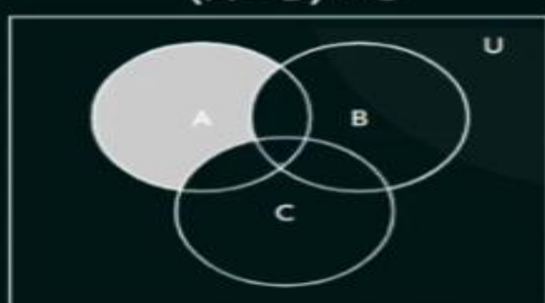


(B - C)

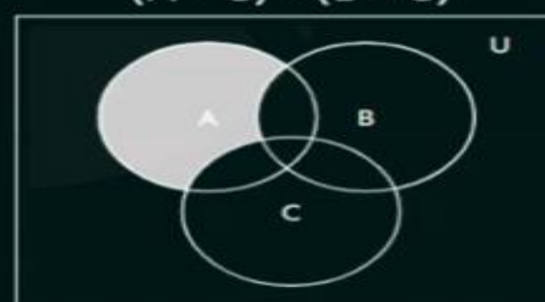
(A - C) - (B - C)



(A - B) - C



(A - C) - (B - C)



Therefore, $X = Y$

Set Identities

Let's consider some basic set identities.
Let say A, B and C are three sets and these three sets are subsets of the universal set U.

Set Identities:

1. Identity Laws:

   a) $A \cup \phi = A$

   b) $A \cap U = A$

Proof: a) $A \cup \phi = A$

$A \cup \phi = \{x \mid x \in A \vee x \in \phi\}$

We know that $x \in \phi$ is false.

$A \cup \phi = \{x \mid x \in A \vee F\}$

$A \cup \phi = \{x \mid x \in A\}$

$A \cup \phi = A$

b) $A \cap U = A$

$A \cap U = \{x \mid x \in A \wedge x \in U\}$

We know that every element is contained in the universal set U,
Therefore, $x \in U$ is always True.

$A \cap U = \{x \mid x \in A \wedge T\}$

$A \cap U = \{x \mid x \in A\}$

$A \cap U = A$

2. Domination Laws:

a) $A \cup U = U$

b) $A \cap \phi = \phi$

Proof: a) $A \cup U = U$

$A \cup U = \{x \mid x \in A \vee x \in U\}$

$A \cup U = \{x \mid x \in A \vee T\}$

$A \cup U = \{x \mid T\}$

We know that $x \in U = T$

So we can replace T by $x \in U$

$A \cup U = \{x \mid x \in U\}$

$A \cup U = U$

b) $A \cup \phi = \phi$

$A \cap \phi = \{x \mid x \in A \wedge x \in \phi\}$

$A \cap \phi = \{x \mid x \in A \wedge F\}$

$A \cap \phi = \{x \mid F\}$

We know that $x \in \phi$ is False (F)

Therefore, F can be replaced by $x \in \phi$.

$A \cap \phi = \{x \mid x \in \phi\}$

$A \cap \phi = \phi$

3. Idempotent Laws:

a) $A \cup A = A$

b) $A \cap A = A$

Proof: a) $A \cup A = A$

$A \cup A = \{x \mid x \in A \vee x \in A\}$

$A \cup A = \{x \mid x \in A\}$

$A \cup A = A$

b) $A \cap A = A$

$A \cap A = \{x \mid x \in A \wedge x \in A\}$

$A \cap A = \{x \mid x \in A\}$

$A \cap A = A$

Set Identities (Part II)

4. Complementation Laws:

$(A')' = A$

Proof: $(A')' = \{x \mid x \notin A'\}$
$(A')' = \{x \mid \neg(x \in A')\}$
$(A')' = \{x \mid \neg(x \notin A)\}$
$(A')' = \{x \mid \neg(\neg(x \in A))\}$
$(A')' = \{x \mid x \in A\}$
$(A')' = A$

5. Commutative Laws:

a) $A \cup B = B \cup A$
b) $A \cap B = B \cap A$

Proof:

a) $A \cup B = \{x \mid x \in A \vee x \in B\}$
$A \cup B = \{x \mid x \in B \vee x \in A\}$
$A \cup B = B \cup A$

b) $A \cap B = \{x \mid x \in A \wedge x \in B\}$
$A \cap B = \{x \mid x \in B \wedge x \in A\}$
$A \cap B = B \cap A$

6. Associative Laws:

a) $A \cup (B \cup C) = (A \cup B) \cup C$
b) $A \cap (B \cap C) = (A \cap B) \cap C$

Proof:

a) $A \cup (B \cup C) = \{x \mid x \in A \vee (x \in B \vee x \in C)\}$
By first associative law of logical equivalences
$p \vee (q \vee r) = (p \vee q) \vee r$

$A \cup (B \cup C) = \{x \mid (x \in A \vee x \in B) \vee x \in C\}$
$A \cup (B \cup C) = (A \cup B) \cup C$

b) $A \cap (B \cap C) = \{x \mid x \in A \wedge (x \in B \wedge x \in C)\}$
By second associative law of logical equivalences
$p \wedge (q \wedge r) = (p \wedge q) \wedge r$

$A \cap (B \cap C) = \{x \mid (x \in A \wedge x \in B) \wedge x \in C\}$
$A \cap (B \cap C) = (A \cap B) \cap C$

7. Distributive Laws:

a) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
b) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Proof: a) $A \cap (B \cup C) = \{x \mid x \in A \wedge x \in (B \cup C)\}$

$A \cap (B \cup C) = \{x \mid x \in A \wedge (x \in B \vee x \in C)\}$

From the rules of logic

$p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$

$A \cap (B \cup C) = \{x \mid (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C)\}$

$A \cap (B \cup C) = \{x \mid x \in (A \cap B) \vee x \in (A \cap C)\}$

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

b) $A \cup (B \cap C) = \{x \mid x \in A \vee (x \in B \wedge x \in C)\}$

From the rules of logic

$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$

$A \cup (B \cap C) = \{x \mid (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\}$

$A \cup (B \cap C) = \{x \mid (x \in (A \cup B) \wedge x \in (A \cup C))\}$

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Set Identities (Part III)

8. De Morgan's Laws:

a) $(A\cup B)' = A'\cap B'$
b) $(A\cap B)' = A'\cup B'$

Proof:

a) $(A\cup B)' = \{x \mid x \notin (A\cup B)\}$
$= \{x \mid \neg(x \in (A\cup B))\}$
$= \{x \mid \neg(x\in A \vee x\in B)\}$
By using first De Morgan's law
$= \{x \mid (x\notin A \wedge x\notin B)\}$
$= \{x \mid (x\in A' \wedge x\in B')\}$
$= \{x \mid (x\in(A'\cap B'))\}$
$= A'\cap B'$

b) $(A\cap B)' = \{x \mid x \notin (A\cap B)\}$
$= \{x \mid \neg(x \in (A\cap B))\}$
$= \{x \mid \neg(x\in A \wedge x\in B)\}$
By using second De Morgan's law of logical equivalences
$= \{x \mid (x\notin A \vee x\notin B)\}$
$= \{x \mid (x\in A' \vee x\in B')\}$
$= \{x \mid (x\in(A'\cup B'))\}$
$= A'\cup B'$

9. Absorption Laws:

a) $A\cup(A\cap B) = A$
b) $A\cap(A\cup B) = A$

Proof: a) $A\cup(A\cap B) = A$
Prove that $A\cup(A\cap B) \subseteq A$ and $A \subseteq A\cup(A\cap B)$

1. $A\cup(A\cap B) \subseteq A$
Let $x\in(A\cup(A\cap B))$

$x\in A \vee x\in A\cap B$
$= x\in A \vee (x\in A \wedge x\in B)$

$= x \in A \vee (x \in A \wedge x \in B)$
By first absorption law of logical equivalences $[p \vee (p \wedge q) = p]$
$= x \in A$
Therefore, $A \cup (A \cap B) \subseteq A$

2. $A \subseteq A \cup (A \cap B)$
Let $x \in A$
We know that an element of $A \cup (A \cap B)$ is an element that is in A or $A \cap B$
Therefore, $x \in A \cup (A \cap B)$
Hence, $A \subseteq A \cup (A \cap B)$

$A = A \cup (A \cap B)$

b) LHS $= A \cap (A \cup B)$
$= \{x \mid x \in A \wedge x \in (A \cup B)\}$
$= \{x \mid x \in A \wedge (x \in A \vee x \in B)\}$
From second absorption law of logical equivalences $[p \wedge (p \vee q) = p]$
$= \{x \mid x \in A\}$
$= A =$ RHS

10. Complement Laws:

a) $A \cup A' = U$
b) $A \cap A' = \phi$

Proof: a) LHS $= A \cup A'$
$= \{x \mid x \in A \vee x \in A'\}$
$= U =$ RHS

b) LHS $= A \cap A'$
$= \{x \mid x \in A \wedge x \in A'\}$
$= \phi =$ RHS

Set Operations (Solved Problem 1)

Problem: Show that if A and B are sets, then

a) $A - B = A \cap B'$ b) $(A \cap B) \cup (A \cap B') = A$

Solution:

a) LHS $= A - B$

$= \{x \mid x \in A \wedge x \notin B\}$

$= \{x \mid x \in A \wedge \neg(x \in B)\}$

$= \{x \mid x \in A \wedge x \in B'\}$

$= \{x \mid x \in (A \cap B')\}$

$= A \cap B' =$ RHS

b) LHS $= (A \cap B) \cup (A \cap B')$

$= \{x \mid x \in (A \cap B) \vee x \in (A \cap B')\}$

$= \{x \mid (x \in A \wedge x \in B) \vee (x \in A \wedge x \in B')\}$

From the distributive law of logical equivalence

$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

$= \{x \mid x \in A \wedge (x \in B \vee x \in B')\}$

$= \{x \mid x \in A \wedge T\}$

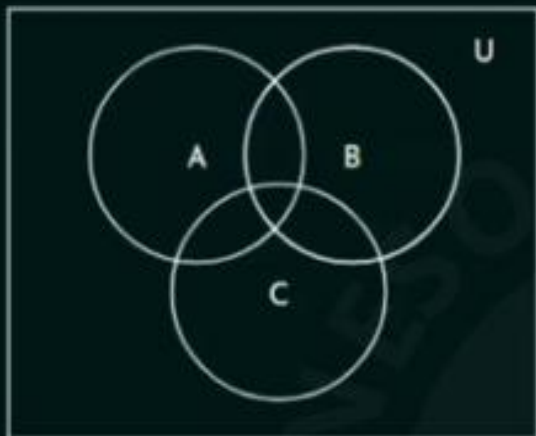
$= \{x \mid x \in A\} = A =$ RHS

Set Operations (Solved Problem 2)

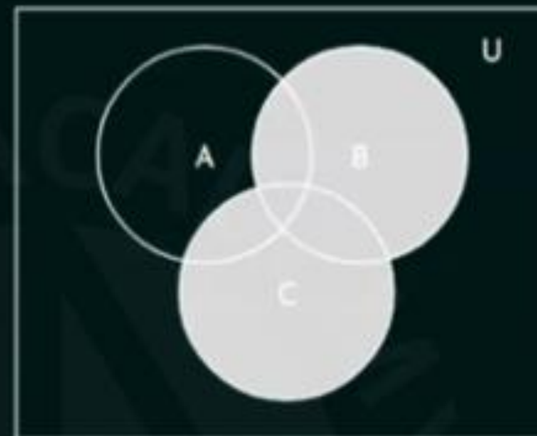
Problem: Draw Venn diagrams for each of these combinations of the sets A, B and C

a) $A \cap (B \cup C)$ b) $A' \cap B' \cap C'$ c) $(A-B) \cup (A-C) \cup (B-C)$

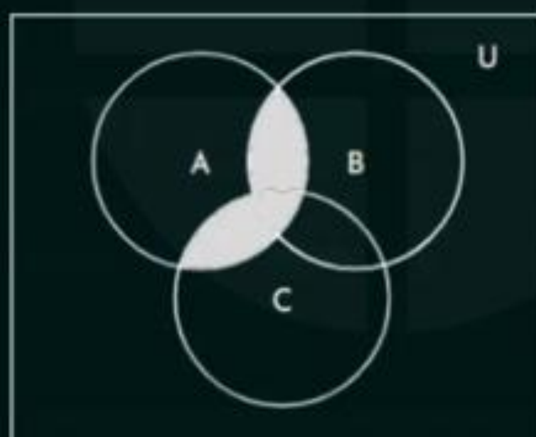
Solution:



A, B and C

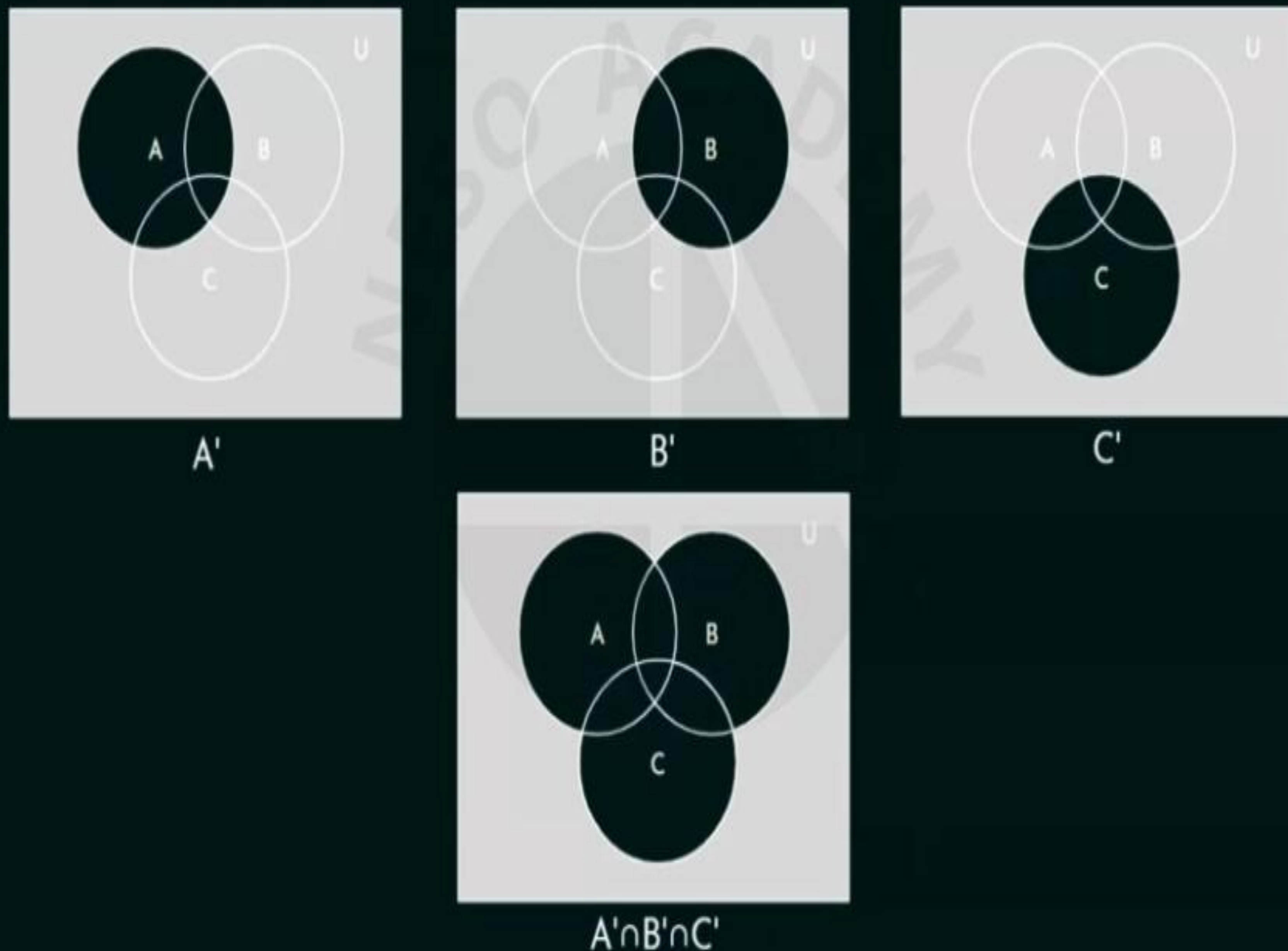


$B \cup C$

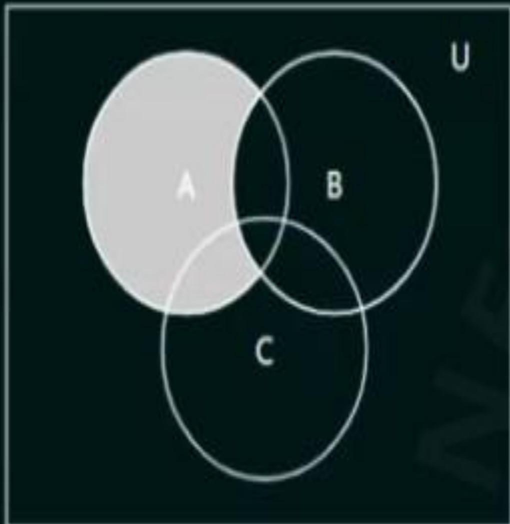


$A \cap (B \cup C)$

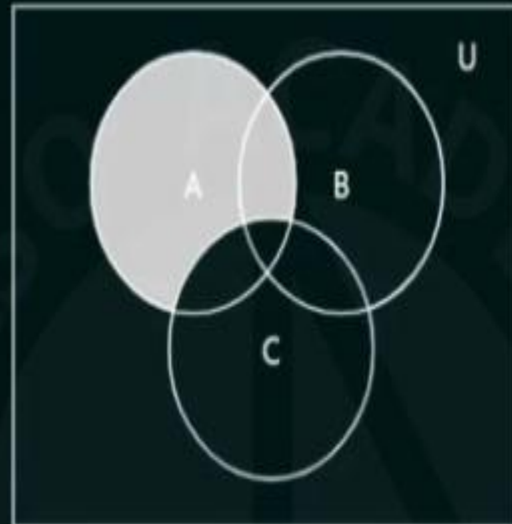
b) $A' \cap B' \cap C'$



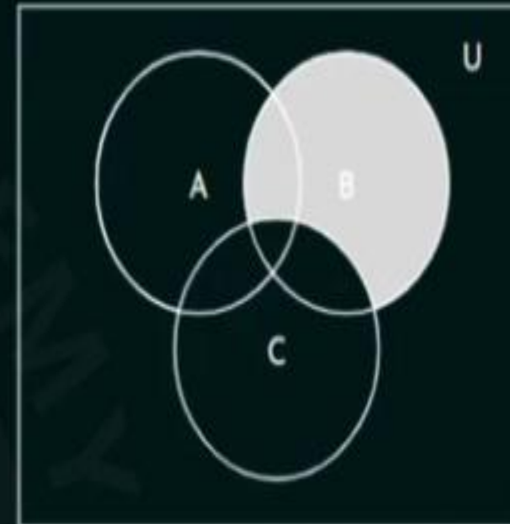
c) $(A-B) \cup (A-C) \cup (B-C)$



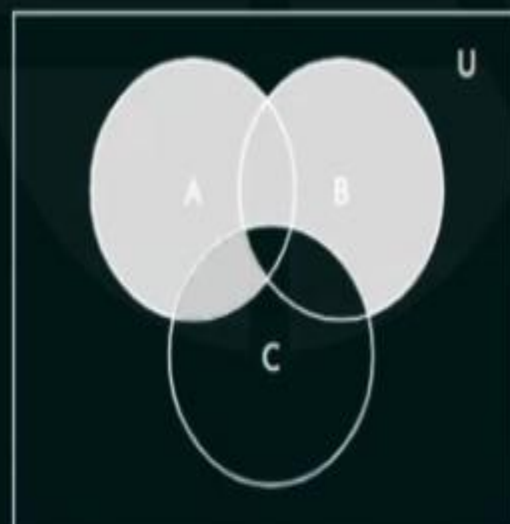
A-B



A-C



B-C



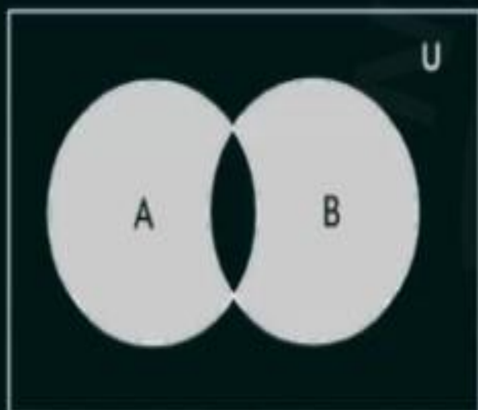
$(A-B) \cup (A-C) \cup (B-C)$

Symmetric Difference

Symmetric difference of sets A and B, denoted by $A \oplus B$, is the set of all elements which are in either set A or set B, but not in both A and B.

$$A \oplus B = \{x \mid (x \in A \lor x \in B) \land \neg(x \in A \land x \in B)\}$$

Venn diagram representation of $A \oplus B$



Note that:
1. $A \oplus B = (A \cup B) - (A \cap B)$
2. $A \oplus B = (A - B) \cup (B - A)$

Proof: 1. $A \oplus B = (A \cup B) - (A \cap B)$

$$\begin{aligned}
\text{LHS} &= A \oplus B \\
&= \{x \mid (x \in A \lor x \in B) \land \neg(x \in A \land x \in B)\} \\
&= \{x \mid (x \in (A \cup B)) \land \neg(x \in A \land x \in B)\} \\
&= \{x \mid (x \in (A \cup B)) \land \neg(x \in (A \cap B))\} \\
&= \{x \mid (x \in ((A \cup B) - (A \cap B))\} \\
&= (A \cup B) - (A \cap B) = \text{RHS}
\end{aligned}$$

2. $A \oplus B = (A-B) \cup (B-A)$

LHS $= A \oplus B$

$= \{x \mid (x \in A \vee x \in B) \wedge \neg(x \in A \wedge x \in B)\}$

$= \{x \mid (x \in A \vee x \in B) \wedge (\neg(x \in A) \vee \neg(x \in B))\}$

$= \{x \mid (x \in A \vee x \in B) \wedge (x \notin A \vee x \notin B)\}$

$= \{x \mid ((x \in A \wedge x \notin A) \vee (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A) \vee (x \in B \wedge x \notin B))\}$

$= \{x \mid (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$

RHS $= (A-B) \cup (B-A)$

$= \{x \mid x \in (A-B) \vee x \in (B-A)\}$

$= \{x \mid (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$

Therefore, LHS = RHS

Problem: Show that if A is a subset of Universal set U, then

a) $A \oplus A = \phi$ b) $A \oplus \phi = A$ c) $A \oplus U = A'$ d) $A \oplus A' = U$

Solution: a) $A \oplus A = \phi$

LHS $= A \oplus A$

$= \{x \mid (x \in A \vee x \in A) \wedge \neg(x \in A \wedge x \in A)\}$

$= \{x \mid x \in A \wedge \neg(x \in A)\}$

$= \{x \mid x \in A \wedge x \notin A\} = \phi =$ RHS

b) $A \oplus \phi = A$

LHS $= A \oplus \phi$

$= \{x \mid (x \in A \vee x \in \phi) \wedge \neg(x \in A \wedge x \in \phi)\}$

$= \{x \mid (x \in A \vee F) \wedge \neg(x \in A \wedge F)\}$

$= \{x \mid x \in A \wedge \neg(F)\}$

$= \{x \mid x \in A \wedge T\}$

$= \{x \mid x \in A\} = A =$ RHS

c) $A \oplus U = A'$

$LHS = A \oplus U$

$= \{x \mid (x \in A \vee x \in U) \wedge \neg(x \in A \wedge x \in U)\}$

$= \{x \mid (x \in A \vee T) \wedge \neg(x \in A \wedge T)\}$

$= \{x \mid T \wedge \neg(x \in A)\}$

$= \{x \mid \neg(x \in A)\}$

$= \{x \mid x \notin A\} = A' = RHS$

d) $A \oplus A' = U$

$LHS = A \oplus A'$

$= \{x \mid (x \in A \vee x \in A') \wedge \neg(x \in A \wedge x \in A')\}$

$= \{x \mid (x \in A \vee x \in A') \wedge (x \notin A \vee x \notin A')\}$

$= \{x \mid (x \in A \vee x \in A') \wedge (x \in A' \vee x \in A)\}$

$= \{x \mid (x \in A \vee x \in A')\} = U = RHS$

Set Operations [GATE Problems]

Problem 1: Let A and B be sets and let A' and B' denote the complement of the sets A and B. The set $(A-B) \cup (B-A) \cup (A \cap B)$ is equal to

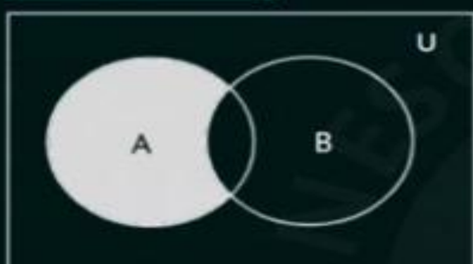
a) $A \cup B$ b) $A' \cup B'$ c) $A \cap B$ d) $A' \cap B'$ [GATE 1996]

Solution: $(A-B) \cup (B-A) \cup (A \cap B)$

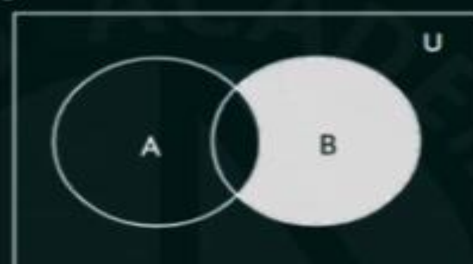
$(A-B)$ means "only A"
$(B-A)$ means "only B"
$(A \cap B)$ means "Both A and B"

(Only A) $\cup$ (Only B) $\cup$ (Both A and B) $= A \cup B$

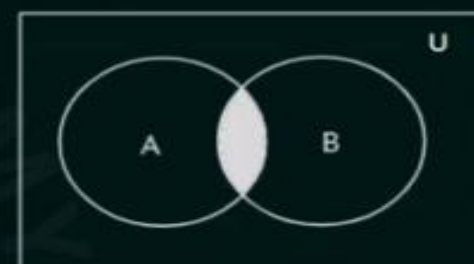
Alternate way: Venn Diagrams



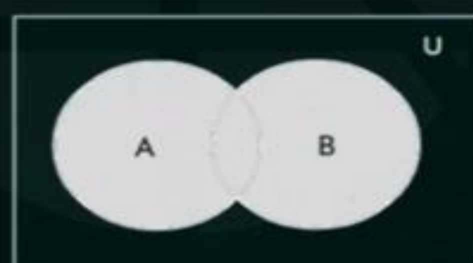
A-B



B-A



$A \cap B$



$(A-B) \cup (B-A) \cup (A \cap B)$

Problem 2: Let E, F and G be the finite sets. Let

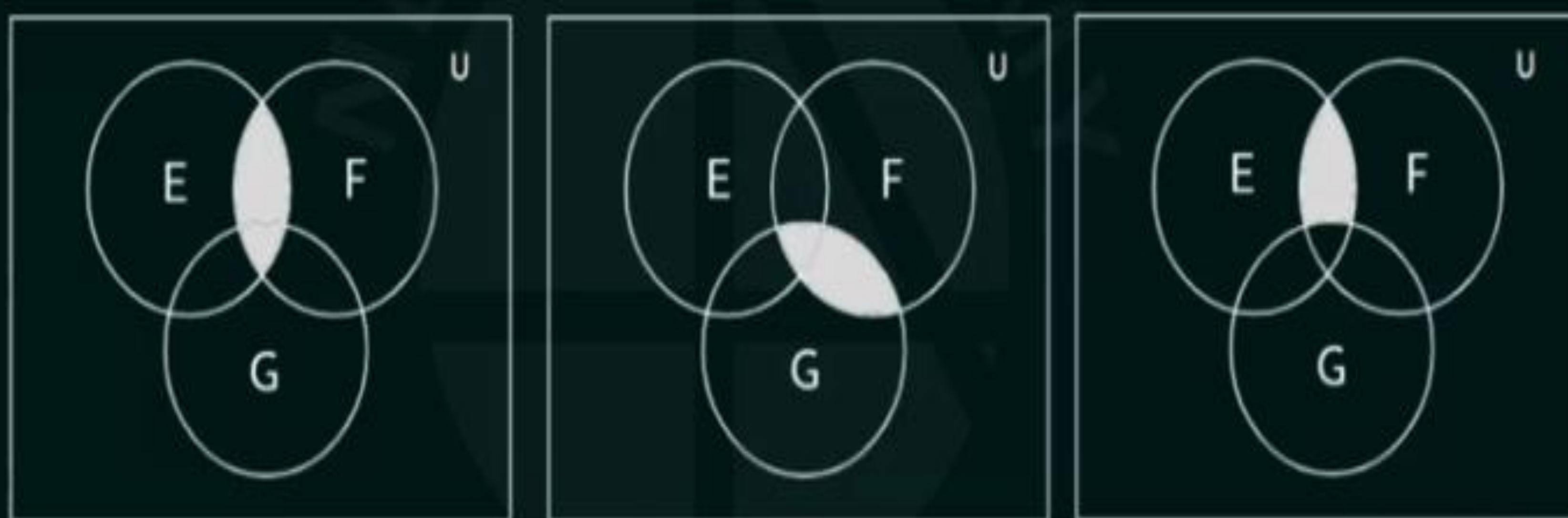
$X = (E \cap F) - (F \cap G)$ and
$Y = (E - (E \cap G)) - (E - F)$

Which one of the following is true?

(A) $X \subset Y$ (C) $X = Y$
(B) $X \supset Y$ (D) $X - Y \neq \phi$ and $Y - X \neq \phi$ [GATE 2006]

Solution: Venn diagram

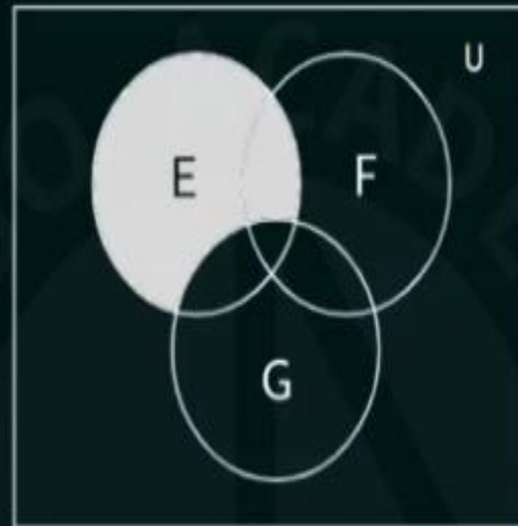
1. $X = (E \cap F) - (F \cap G)$



$E \cap F$ $F \cap G$ $(E \cap F) - (F \cap G)$

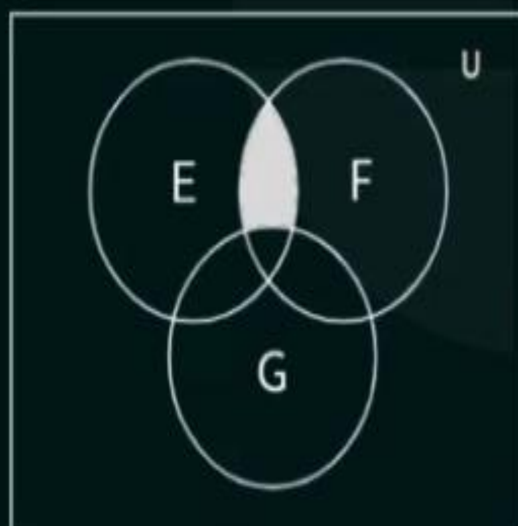
2. $(E - (E \cap G)) - (E - F)$

$E \cap G$

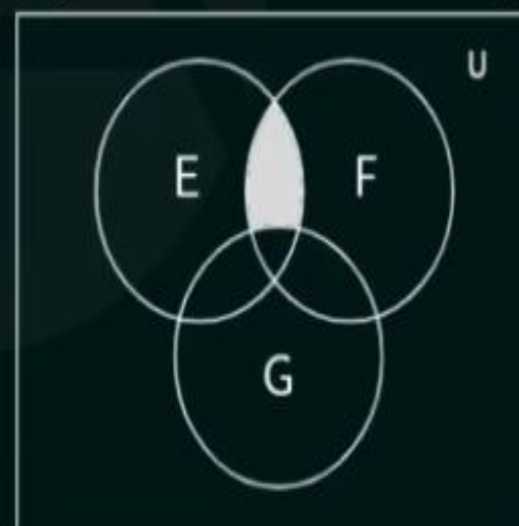


$E - (E \cap G)$

$E - F$



$(E - (E \cap G)) - (E - F)$



$(E \cap F) - (F \cap G)$

$X = Y$

Problem:

Let:

- $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ (Universal Set)
- $A = \{2, 4, 6, 8\}$
- $B = \{1, 2, 3, 4\}$
- $C = \{5, 6, 7\}$

Solve the following step-by-step:

1. $(A \cap B) \cup (B \cap C)$
2. $(A \cup B) \cap (B^c \cup C)$
3. $(A \cap C^c) \cup (B^c \cap C)$

Solution:

1. $(A \cap B) \cup (B \cap C)$

- Step 1: Find $A \cap B$:
  $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
  $A = \{2, 4, 6, 8\}, B = \{1, 2, 3, 4\}$
  $A \cap B = \{2, 4\}$
- Step 2: Find $B \cap C$:
  $B \cap C = \{x \mid x \in B \text{ and } x \in C\}$
  $B = \{1, 2, 3, 4\}, C = \{5, 6, 7\}$
  $B \cap C = \emptyset$
- Step 3: Find $(A \cap B) \cup (B \cap C)$:
  $(A \cap B) \cup (B \cap C) = \{2, 4\} \cup \emptyset = \{2, 4\}$

2. $(A \cup B) \cap (B^c \cup C)$

- Step 1: Find $A \cup B$:
  $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
  $A = \{2, 4, 6, 8\}, B = \{1, 2, 3, 4\}$
  $A \cup B = \{1, 2, 3, 4, 6, 8\}$
- Step 2: Find B^c:
  $B^c = \{x \mid x \notin B, x \in U\}$
  $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}, B = \{1, 2, 3, 4\}$
  $B^c = \{5, 6, 7, 8, 9, 10\}$
- Step 3: Find $B^c \cup C$:
  $B^c \cup C = \{x \mid x \in B^c \text{ or } x \in C\}$
  $B^c = \{5, 6, 7, 8, 9, 10\}, C = \{5, 6, 7\}$
  $B^c \cup C = \{5, 6, 7, 8, 9, 10\}$
- Step 4: Find $(A \cup B) \cap (B^c \cup C)$:
  $(A \cup B) \cap (B^c \cup C) = \{x \mid x \in (A \cup B) \text{ and } x \in (B^c \cup C)\}$
  $(A \cup B) = \{1, 2, 3, 4, 6, 8\}, (B^c \cup C) = \{5, 6, 7, 8, 9, 10\}$
  $(A \cup B) \cap (B^c \cup C) = \{6, 8\}$

3. $(A \cap C^c) \cup (B^c \cap C)$

- **Step 1:** Find C^c:
  $C^c = \{x \mid x \notin C, x \in U\}$
  $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}, C = \{5, 6, 7\}$
  $C^c = \{1, 2, 3, 4, 8, 9, 10\}$
- **Step 2:** Find $A \cap C^c$:
  $A \cap C^c = \{x \mid x \in A \text{ and } x \in C^c\}$
  $A = \{2, 4, 6, 8\}, C^c = \{1, 2, 3, 4, 8, 9, 10\}$
  $A \cap C^c = \{2, 4, 8\}$
- **Step 3:** Find $B^c \cap C$:
  $B^c \cap C = \{x \mid x \in B^c \text{ and } x \in C\}$
  $B^c = \{5, 6, 7, 8, 9, 10\}, C = \{5, 6, 7\}$
  $B^c \cap C = \{5, 6, 7\}$
- **Step 4:** Find $(A \cap C^c) \cup (B^c \cap C)$:
  $(A \cap C^c) \cup (B^c \cap C) = \{2, 4, 8\} \cup \{5, 6, 7\}$
  $= \{2, 4, 5, 6, 7, 8\}$

Exercises

1. List the members of these sets.

a) $\{x \mid x$ is a real number such that $x^2 = 1\}$

b) $\{x \mid x$ is a positive integer less than $12\}$

c) $\{x \mid x$ is the square of an integer and $x < 100\}$

d) $\{x \mid x$ is an integer such that $x^2 = 2\}$

2. Use set builder notation to give a description of each of these sets.

a) $\{0, 3, 6, 9, 12\}$

b) $\{-3, -2, -1, 0, 1, 2, 3\}$

c) $\{m, n, o, p\}$

3. For each of these pairs of sets, determine whether the first is a subset of the second, the second is a subset of the first, or neither is a subset of the other.

a) the set of airline flights from New York to New Delhi, the set of nonstop airline flights from New York to New Delhi

b) the set of people who speak English, the set of people who speak Chinese

c) the set of flying squirrels, the set of living creatures that can fly

4. For each of these pairs of sets, determine whether the first is a subset of the second, the second is a subset of the first, or neither is a subset of the other.

a) the set of people who speak English, the set of people who speak English with an Australian accent

b) the set of fruits, the set of citrus fruits

c) the set of students studying discrete mathematics, the set of students studying data structures

5. Determine whether each of these pairs of sets are equal.

a) $\{1, 3, 3, 3, 5, 5, 5, 5, 5\}, \{5, 3, 1\}$

b) $\{\{1\}\}, \{1, \{1\}\}$ **c)** $\emptyset, \{\emptyset\}$

6. Suppose that $A = \{2, 4, 6\}$, $B = \{2, 6\}$, $C = \{4, 6\}$, and $D = \{4, 6, 8\}$. Determine which of these sets are subsets of which other of these sets.

7. For each of the following sets, determine whether 2 is an element of that set.

a) $\{x \in \mathbf{R} \mid x$ is an integer greater than $1\}$

b) $\{x \in \mathbf{R} \mid x$ is the square of an integer$\}$

c) $\{2, \{2\}\}$ **d)** $\{\{2\}, \{\{2\}\}\}$

e) $\{\{2\}, \{2, \{2\}\}\}$ **f)** $\{\{\{2\}\}\}$

8. For each of the sets in Exercise 7, determine whether $\{2\}$ is an element of that set.

9. Determine whether each of these statements is true or false.

a) $0 \in \emptyset$ **b)** $\emptyset \in \{0\}$

c) $\{0\} \subset \emptyset$ **d)** $\emptyset \subset \{0\}$

e) $\{0\} \in \{0\}$ **f)** $\{0\} \subset \{0\}$

g) $\{\emptyset\} \subseteq \{\emptyset\}$

10. Determine whether these statements are true or false.

a) $\emptyset \in \{\emptyset\}$ **b)** $\emptyset \in \{\emptyset, \{\emptyset\}\}$

c) $\{\emptyset\} \in \{\emptyset\}$ **d)** $\{\emptyset\} \in \{\{\emptyset\}\}$

e) $\{\emptyset\} \subset \{\emptyset, \{\emptyset\}\}$ **f)** $\{\{\emptyset\}\} \subset \{\emptyset, \{\emptyset\}\}$

g) $\{\{\emptyset\}\} \subset \{\{\emptyset\}, \{\emptyset\}\}$

11. Determine whether each of these statements is true or false.

a) $x \in \{x\}$ **b)** $\{x\} \subseteq \{x\}$ **c)** $\{x\} \in \{x\}$

d) $\{x\} \in \{\{x\}\}$ **e)** $\emptyset \subseteq \{x\}$ **f)** $\emptyset \in \{x\}$

12. Use a Venn diagram to illustrate the subset of odd integers in the set of all positive integers not exceeding 10.

13. Use a Venn diagram to illustrate the set of all months of the year whose names do not contain the letter R in the set of all months of the year.

14. Use a Venn diagram to illustrate the relationship $A \subseteq B$ and $B \subseteq C$.

15. Use a Venn diagram to illustrate the relationships $A \subset B$ and $B \subset C$.

16. Use a Venn diagram to illustrate the relationships $A \subset B$ and $A \subset C$.

17. Suppose that A, B, and C are sets such that $A \subseteq B$ and $B \subseteq C$. Show that $A \subseteq C$.

18. Find two sets A and B such that $A \in B$ and $A \subseteq B$.

19. What is the cardinality of each of these sets?

a) $\{a\}$ b) $\{\{a\}\}$
c) $\{a, \{a\}\}$ d) $\{a, \{a\}, \{a, \{a\}\}\}$

20. What is the cardinality of each of these sets?

a) $\emptyset$ b) $\{\emptyset\}$
c) $\{\emptyset, \{\emptyset\}\}$ d) $\{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$

21. Find the power set of each of these sets, where a and b are distinct elements.

a) $\{a\}$ b) $\{a, b\}$ c) $\{\emptyset, \{\emptyset\}\}$

22. Can you conclude that $A = B$ if A and B are two sets with the same power set?

23. How many elements does each of these sets have where a and b are distinct elements?

a) $\mathcal{P}(\{a, b, \{a, b\}\})$
b) $\mathcal{P}(\{\emptyset, a, \{a\}, \{\{a\}\}\})$
c) $\mathcal{P}(\mathcal{P}(\emptyset))$

24. Determine whether each of these sets is the power set of a set, where a and b are distinct elements.

a) $\emptyset$ b) $\{\emptyset, \{a\}\}$
c) $\{\emptyset, \{a\}, \{\emptyset, a\}\}$ d) $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

25. Prove that $\mathcal{P}(A) \subseteq \mathcal{P}(B)$ if and only if $A \subseteq B$.

26. Show that if $A \subseteq C$ and $B \subseteq D$, then $A \times B \subseteq C \times D$

27. Let $A = \{a, b, c, d\}$ and $B = \{y, z\}$. Find

a) $A \times B$. b) $B \times A$.

33. Find A^2 if

a) $A = \{0, 1, 3\}$. b) $A = \{1, 2, a, b\}$.

34. Find A^3 if

a) $A = \{a\}$. b) $A = \{0, a\}$.

35. How many different elements does $A \times B$ have if A has m elements and B has n elements?

36. How many different elements does $A \times B \times C$ have if A has m elements, B has n elements, and C has p elements?

37. How many different elements does A^n have when A has m elements and n is a positive integer?

38. Show that $A \times B \neq B \times A$, when A and B are nonempty, unless $A = B$.

39. Explain why $A \times B \times C$ and $(A \times B) \times C$ are not the same.

40. Explain why $(A \times B) \times (C \times D)$ and $A \times (B \times C) \times D$ are not the same.

41. Translate each of these quantifications into English and determine its truth value.

a) $\forall x \in \mathbf{R}\ (x^2 \neq -1)$ b) $\exists x \in \mathbf{Z}\ (x^2 = 2)$
c) $\forall x \in \mathbf{Z}\ (x^2 > 0)$ d) $\exists x \in \mathbf{R}\ (x^2 = x)$

42. Translate each of these quantifications into English and determine its truth value.

a) $\exists x \in \mathbf{R}\ (x^3 = -1)$ b) $\exists x \in \mathbf{Z}\ (x + 1 > x)$
c) $\forall x \in \mathbf{Z}\ (x - 1 \in \mathbf{Z})$ d) $\forall x \in \mathbf{Z}\ (x^2 \in \mathbf{Z})$

43. Find the truth set of each of these predicates where the domain is the set of integers.

a) $P(x)$: $x^2 < 3$ b) $Q(x)$: $x^2 > x$
c) $R(x)$: $2x + 1 = 0$

44. Find the truth set of each of these predicates where the domain is the set of integers.

a) $P(x)$: $x^3 \geq 1$ b) $Q(x)$: $x^2 = 2$
c) $R(x)$: $x < x^2$

*45. The defining property of an ordered pair is that two ordered pairs are equal if and only if their first elements are equal and their second elements are equal. Surprisingly, instead of taking the ordered pair as a primitive concept, we can construct ordered pairs using basic notions

Exercises

1. Let A be the set of students who live within one mile of school and let B be the set of students who walk to classes. Describe the students in each of these sets.
   a) $A \cap B$ b) $A \cup B$
   c) $A - B$ d) $B - A$
2. Suppose that A is the set of sophomores at your school and B is the set of students in discrete mathematics at your school. Express each of these sets in terms of A and B.
   a) the set of sophomores taking discrete mathematics in your school
   b) the set of sophomores at your school who are not taking discrete mathematics
   c) the set of students at your school who either are sophomores or are taking discrete mathematics
   d) the set of students at your school who either are not sophomores or are not taking discrete mathematics
3. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find
   a) $A \cup B$. b) $A \cap B$.
   c) $A - B$. d) $B - A$.
4. Let $A = \{a, b, c, d, e\}$ and $B = \{a, b, c, d, e, f, g, h\}$. Find
   a) $A \cup B$. b) $A \cap B$.
   c) $A - B$. d) $B - A$.

In Exercises 5–10 assume that A is a subset of some underlying universal set U.

5. Prove the complementation law in Table 1 by showing that $\overline{\overline{A}} = A$.
6. Prove the identity laws in Table 1 by showing that
   a) $A \cup \emptyset = A$. b) $A \cap U = A$.
7. Prove the domination laws in Table 1 by showing that
   a) $A \cup U = U$. b) $A \cap \emptyset = \emptyset$.
8. Prove the idempotent laws in Table 1 by showing that
   a) $A \cup A = A$. b) $A \cap A = A$.
9. Prove the complement laws in Table 1 by showing that
   a) $A \cup \overline{A} = U$. b) $A \cap \overline{A} = \emptyset$.
10. Show that
    a) $A - \emptyset = A$. b) $\emptyset - A = \emptyset$.
11. Let A and B be sets. Prove the commutative laws from Table 1 by showing that
    a) $A \cup B = B \cup A$.
    b) using a membership table.
16. Let A and B be sets. Show that
    a) $(A \cap B) \subseteq A$. b) $A \subseteq (A \cup B)$.
    c) $A - B \subseteq A$. d) $A \cap (B - A) = \emptyset$.
    e) $A \cup (B - A) = A \cup B$.
17. Show that if A, B, and C are sets, then $\overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}$
    a) by showing each side is a subset of the other side.
    b) using a membership table.
18. Let A, B, and C be sets. Show that
    a) $(A \cup B) \subseteq (A \cup B \cup C)$.
    b) $(A \cap B \cap C) \subseteq (A \cap B)$.
    c) $(A - B) - C \subseteq A - C$.
    d) $(A - C) \cap (C - B) = \emptyset$.
    e) $(B - A) \cup (C - A) = (B \cup C) - A$.
19. Show that if A and B are sets, then
    a) $A - B = A \cap \overline{B}$.
    b) $(A \cap B) \cup (A \cap \overline{B}) = A$.
20. Show that if A and B are sets with $A \subseteq B$, then
    a) $A \cup B = B$.
    b) $A \cap B = A$.
21. Prove the first associative law from Table 1 by showing that if A, B, and C are sets, then $A \cup (B \cup C) = (A \cup B) \cup C$.
22. Prove the second associative law from Table 1 by showing that if A, B, and C are sets, then $A \cap (B \cap C) = (A \cap B) \cap C$.
23. Prove the first distributive law from Table 1 by showing that if A, B, and C are sets, then $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
24. Let A, B, and C be sets. Show that $(A - B) - C = (A - C) - (B - C)$.
25. Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8, 9, 10\}$. Find
    a) $A \cap B \cap C$. b) $A \cup B \cup C$.
    c) $(A \cup B) \cap C$. d) $(A \cap B) \cup C$.
26. Draw the Venn diagrams for each of these combinations of the sets A, B, and C.
    a) $A \cap (B \cup C)$ b) $\overline{A} \cap \overline{B} \cap \overline{C}$
    c) $(A - B) \cup (A - C) \cup (B - C)$
27. Draw the Venn diagrams for each of these combinations of the sets A, B, and C.

b) $A \cap B = B \cap A$.

12. Prove the first absorption law from Table 1 by showing that if A and B are sets, then $A \cup (A \cap B) = A$.

13. Prove the second absorption law from Table 1 by showing that if A and B are sets, then $A \cap (A \cup B) = A$.

14. Find the sets A and B if $A - B = \{1, 5, 7, 8\}$, $B - A = \{2, 10\}$, and $A \cap B = \{3, 6, 9\}$.

15. Prove the second De Morgan law in Table 1 by showing that if A and B are sets, then $\overline{A \cup B} = \overline{A} \cap \overline{B}$

a) by showing each side is a subset of the other side.

a) $A \cap (B - C)$ **b)** $(A \cap B) \cup (A \cap C)$

c) $(A \cap \overline{B}) \cup (A \cap \overline{C})$

28. Draw the Venn diagrams for each of these combinations of the sets A, B, C, and D.

a) $(A \cap B) \cup (C \cap D)$ **b)** $\overline{A} \cup \overline{B} \cup \overline{C} \cup \overline{D}$

c) $A - (B \cap C \cap D)$

29. What can you say about the sets A and B if we know that

a) $A \cup B = A$? **b)** $A \cap B = A$?

c) $A - B = A$? **d)** $A \cap B = B \cap A$?

e) $A - B = B - A$?

30. Can you conclude that $A = B$ if A, B, and C are sets such that

a) $A \cup C = B \cup C$? **b)** $A \cap C = B \cap C$?

c) $A \cup C = B \cup C$ and $A \cap C = B \cap C$?

31. Let A and B be subsets of a universal set U. Show that $A \subseteq B$ if and only if $\overline{B} \subseteq \overline{A}$.

The **symmetric difference** of A and B, denoted by $A \oplus B$, is the set containing those elements in either A or B, but not in both A and B.

32. Find the symmetric difference of $\{1, 3, 5\}$ and $\{1, 2, 3\}$.

33. Find the symmetric difference of the set of computer science majors at a school and the set of mathematics majors at this school.

34. Draw a Venn diagram for the symmetric difference of the sets A and B.

35. Show that $A \oplus B = (A \cup B) - (A \cap B)$.

36. Show that $A \oplus B = (A - B) \cup (B - A)$.

37. Show that if A is a subset of a universal set U, then

a) $A \oplus A = \emptyset$. **b)** $A \oplus \emptyset = A$.

c) $A \oplus U = \overline{A}$. **d)** $A \oplus \overline{A} = U$.

49. Let A_i be the set of all nonempty bit strings (that is, bit strings of length at least one) of length not exceeding i. Find

a) $\bigcup_{i=1}^{n} A_i$. **b)** $\bigcap_{i=1}^{n} A_i$.

50. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i,

a) $A_i = \{i, i + 1, i + 2, \ldots\}$.

b) $A_i = \{0, i\}$.

c) $A_i = (0, i)$, that is, the set of real numbers x with $0 < x < i$.

d) $A_i = (i, \infty)$, that is, the set of real numbers x with $x > i$.

51. Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$ if for every positive integer i,

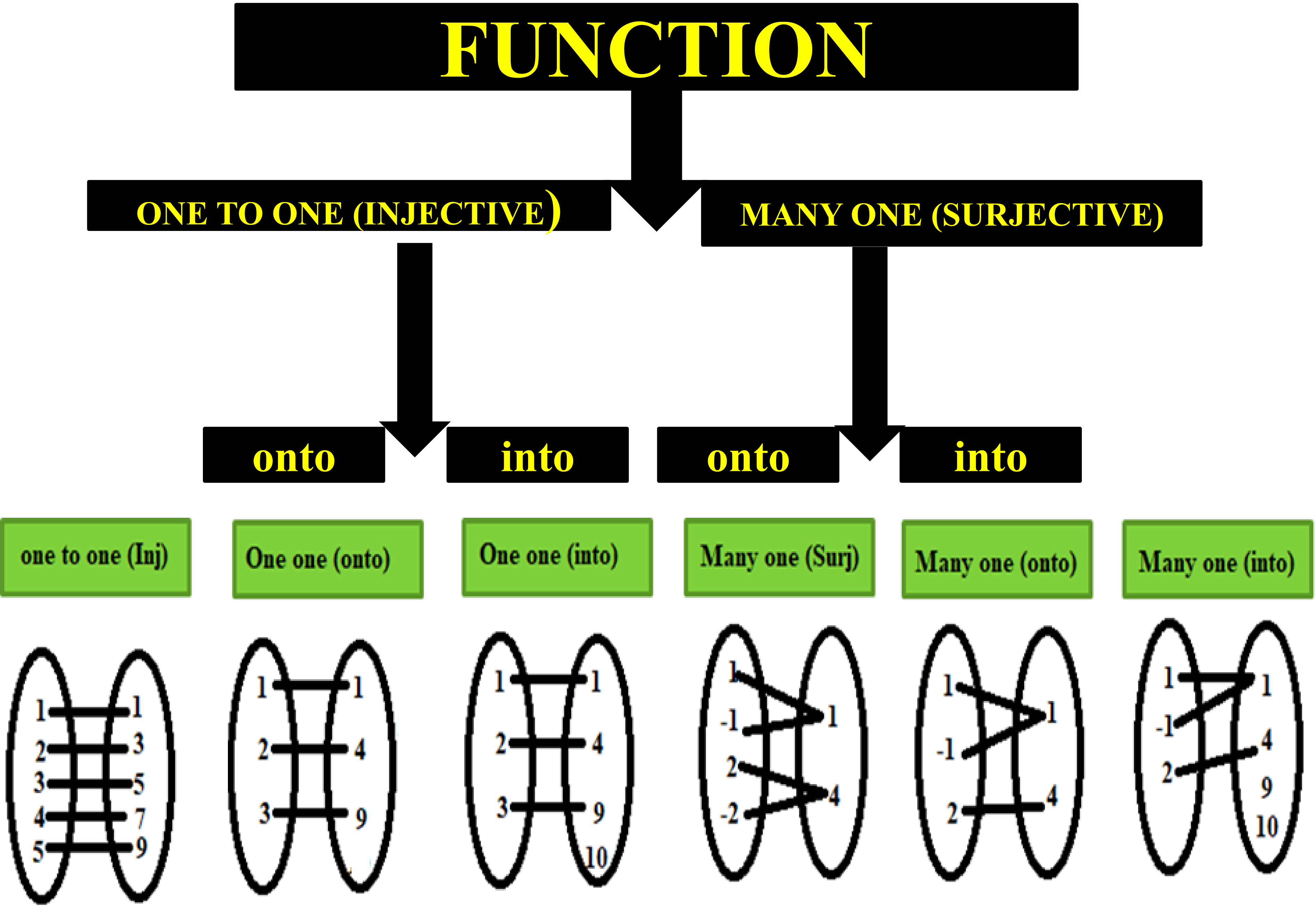
a) $A_i = \{-i, -i + 1, \ldots, -1, 0, 1, \ldots, i - 1, i\}$.

b) $A_i = \{-i, i\}$.

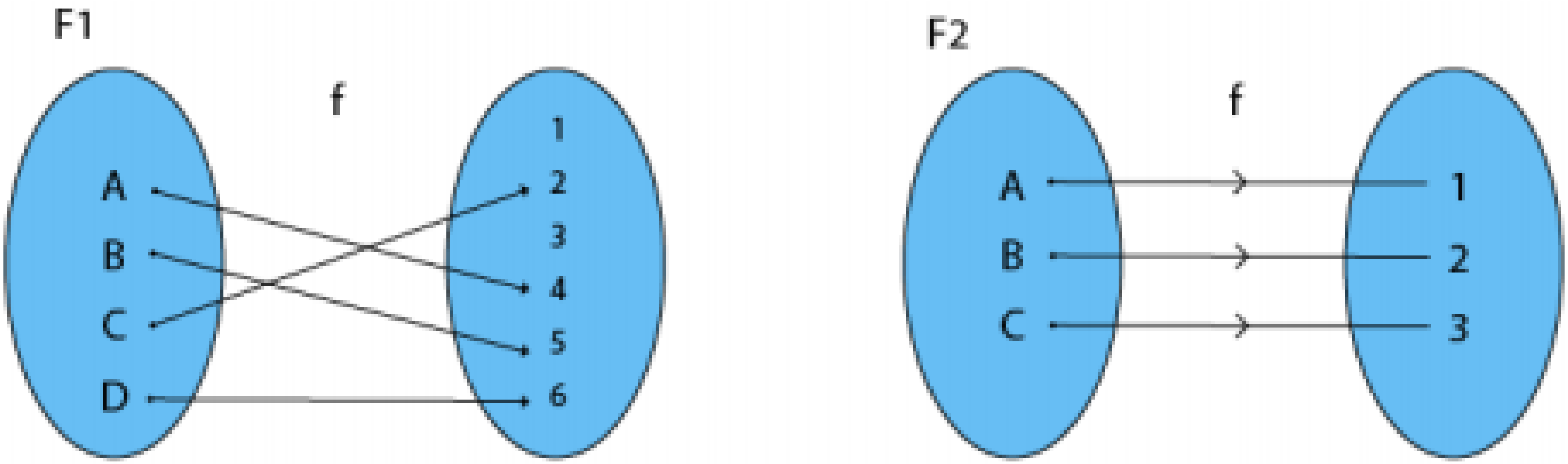
c) $A_i = [-i, i]$, that is, the set of real numbers x with $-i \le x \le i$.

d) $A_i = [i, \infty)$, that is, the set of real numbers x with $x \ge i$.

52. Suppose that the universal set is $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Express each of these sets with bit


FUNCTION
ONE TO ONE (INJECTIVE)
MANY ONE (SURJECTIVE)
onto
into
onto
into
one to one (Inj)
One one (onto)
One one (into)
Many one (Surj)
Many one (onto)
Many one (into)
1
2
3
4
5
1
3
5
7
9
1
2
3
1
4
9
1
2
3
1
4
9
10
1
-1
2
-2
1
4
1
-1
2
1
4
1
-1
2
1
4
9
10

1. Injective (One-to-One) Functions: A function in which one element of Domain Set is connected to one element of Co-Domain Set.

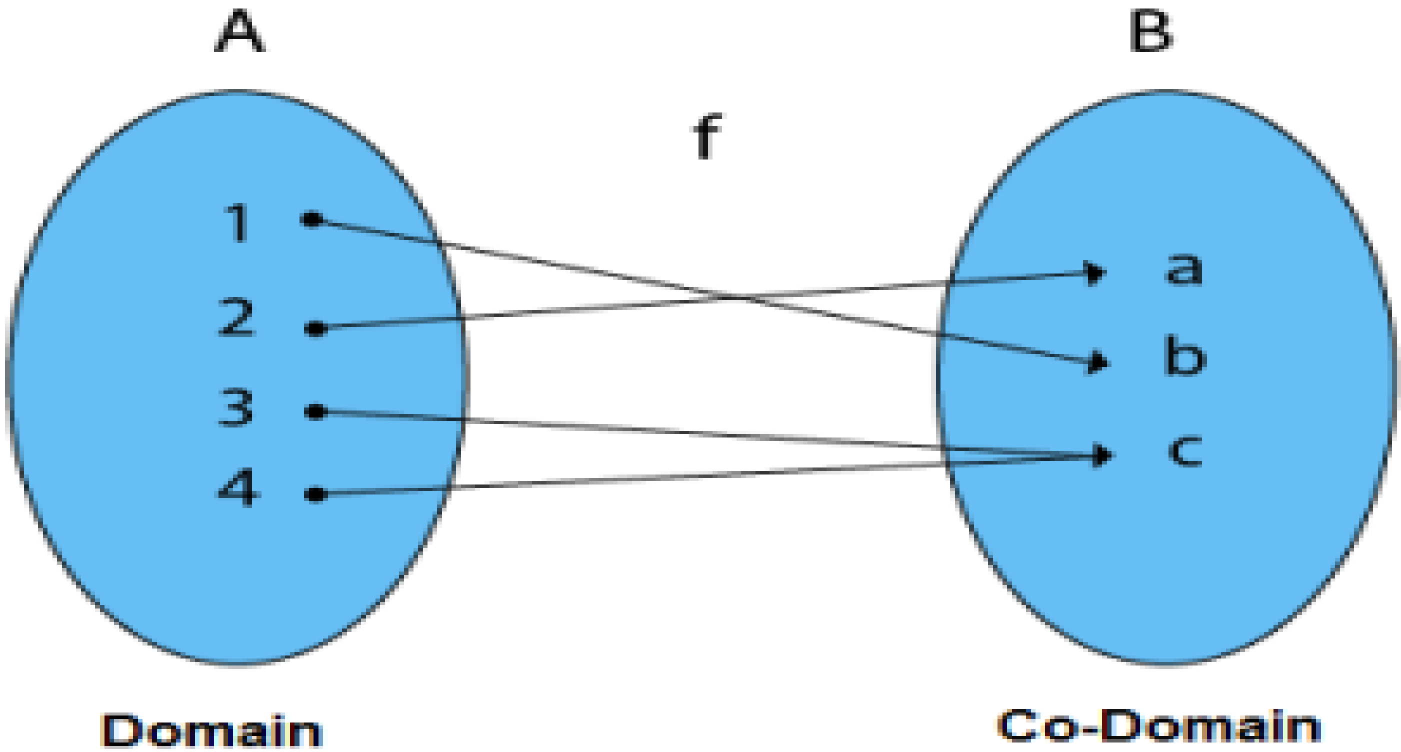


F1 and F2 show one to one Function

2. Surjective (Onto) Functions: A function in which every element of Co-Domain Set has one pre-image.

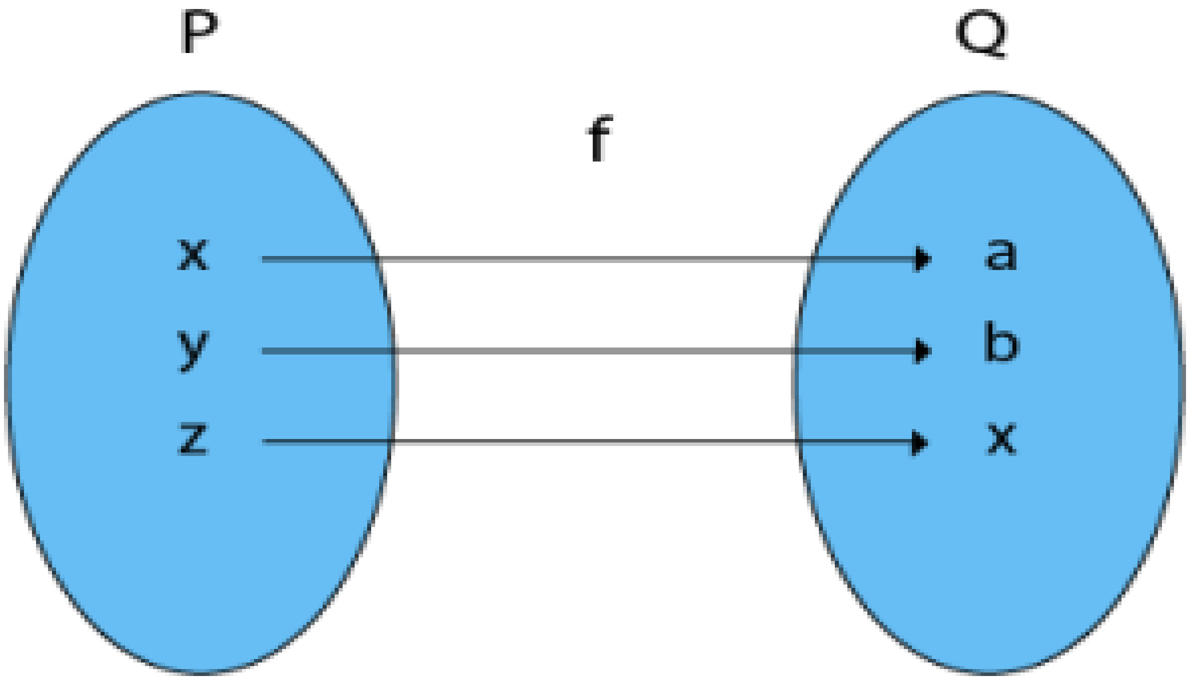
Example: Consider, A = {1, 2, 3, 4}, B = {a, b, c} and f = {(1, b), (2, a), (3, c), (4, c)}.

It is a Surjective Function, as every element of B is the image of some A



Note: In an Onto Function, Range is equal to Co-Domain.

3. Bijective (One-to-One Onto) Functions: A function which is both injective (one to - one) and surjective (onto) is called bijective (One-to-One Onto) Function.



Example:

> Consider P = {x, y, z}
>
> Q = {a, b, c}
>
> and f: P → Q such that
>
> f = {(x, a), (y, b), (z, c)}

The f is a one-to-one function and also it is onto. So it is a bijective function.

4. Into Functions: A function in which there must be an element of co-domain Y does not have a pre-image in domain X.

Example:

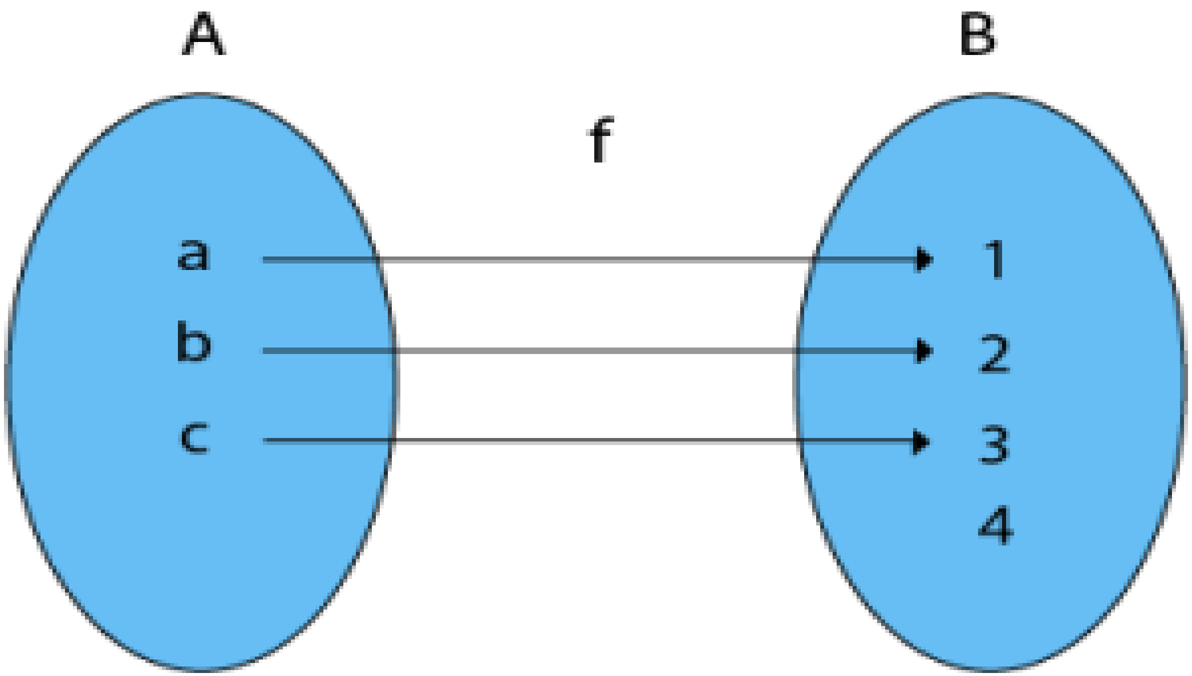
Consider, A = {a, b, c}

B = {1, 2, 3, 4} and f: A → B such that

f = {(a, 1), (b, 2), (c, 3)}

In the function f, the range i.e., {1, 2, 3} ≠ co-domain of Y i.e., {1, 2, 3, 4}

Therefore, it is an into function



5. One-One Into Functions: Let f: X → Y. The function f is called one-one into function if different elements of X have different unique images of Y.

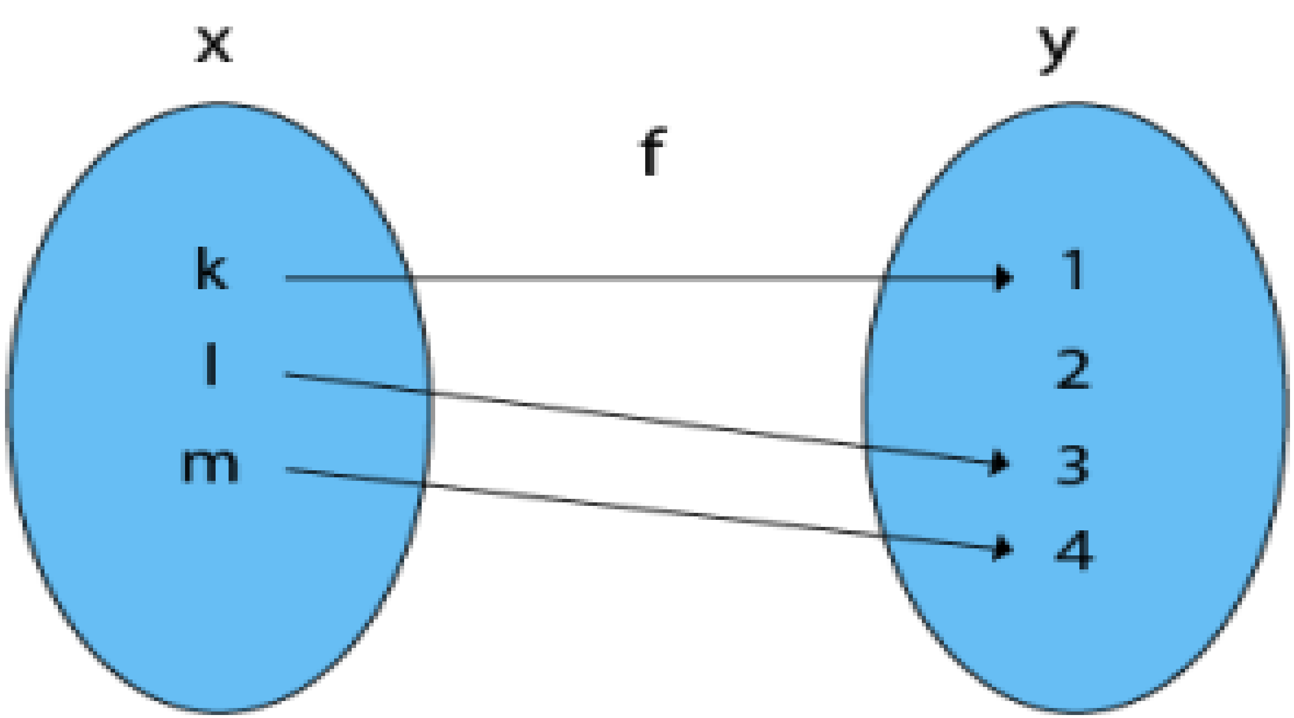
Example:

Consider, X = {k, l, m}

Y = {1, 2, 3, 4} and f: X → Y such that

f = {(k, 1), (l, 3), (m, 4)}

The function f is a one-one into function

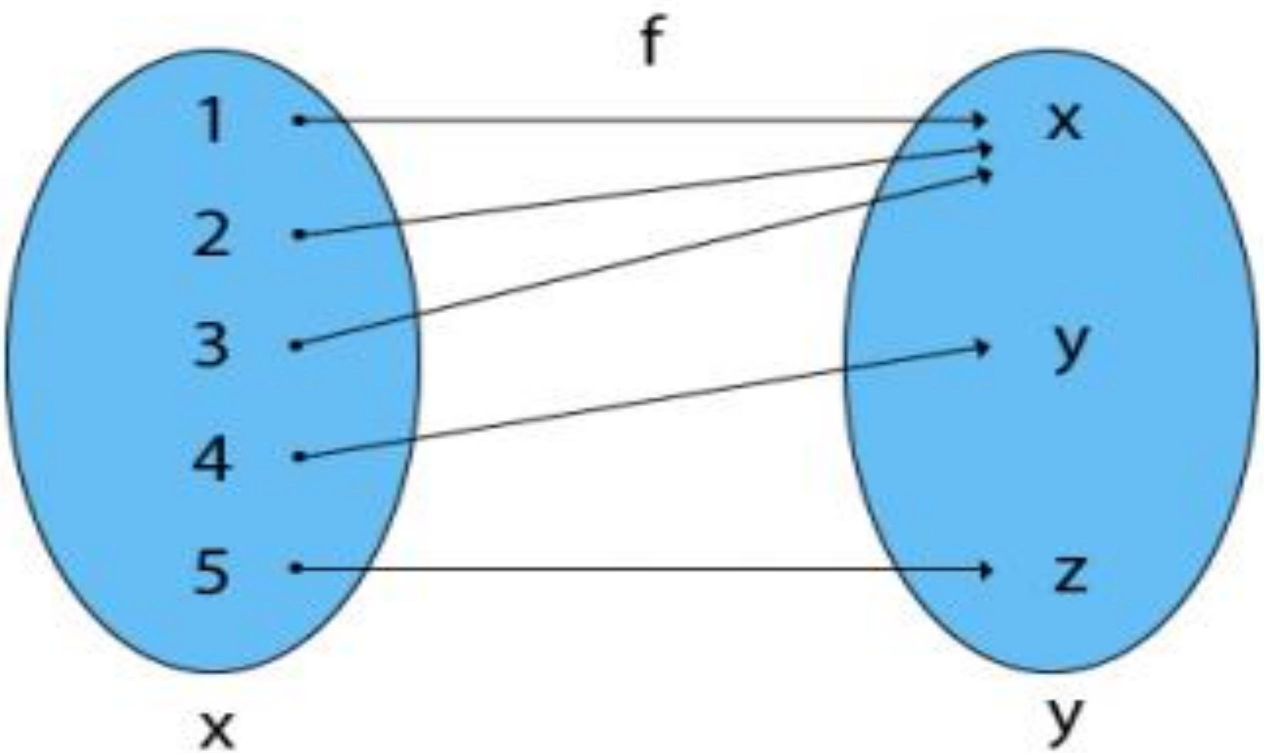


6. Many-One Functions: Let f: X → Y. The function f is said to be many-one functions if there exist two or more than two different elements in X having the same image in Y.

Example:

> Consider X = {1, 2, 3, 4, 5}
>
> Y = {x, y, z} and f: X → Y such that
>
> f = {(1, x), (2, x), (3, x), (4, y), (5, z)}

The function f is a many-one function



7. Many-One Into Functions: Let f: X → Y. The function f is called the many-one function if and only if is both many one and into function.

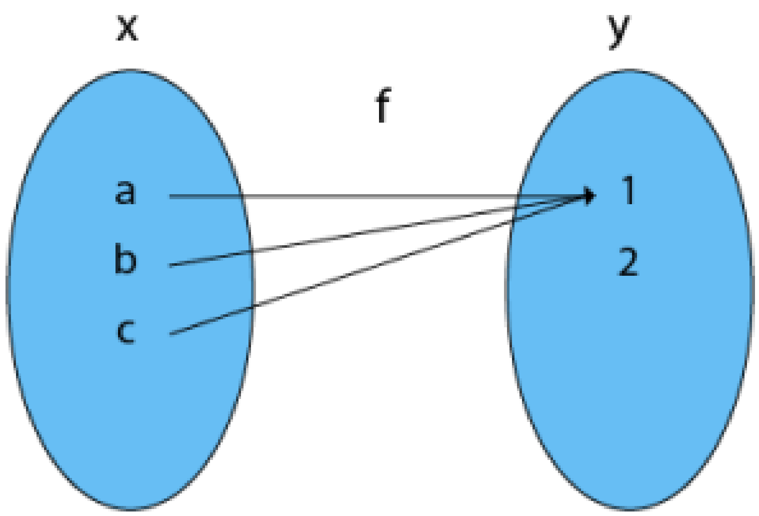
Example:

Consider X = {a, b, c}

Y = {1, 2} and f: X → Y such that

f = {(a, 1), (b, 1), (c, 1)}

As the function f is a many-one and into, so it is a many-one into function.

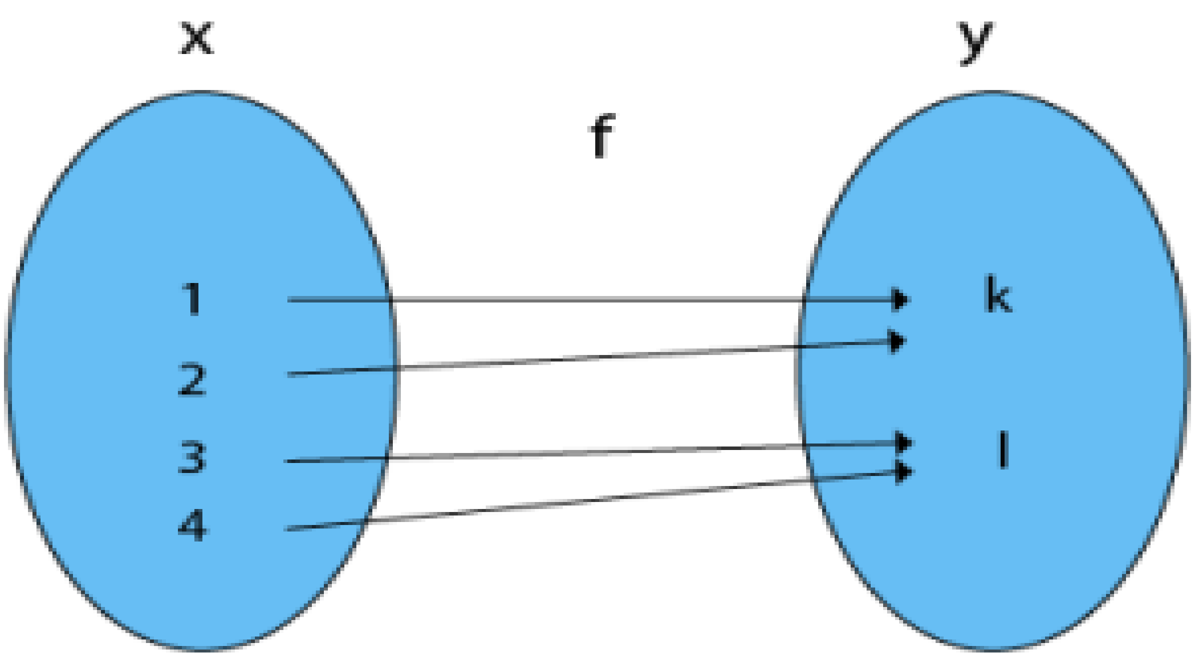


8. Many-One Onto Functions: Let f: X → Y. The function f is called many-one onto function if and only if is both many one and onto.

Example:

> Consider X = {1, 2, 3, 4}
>
> Y = {k, l} and f: X → Y such that
>
> f = {(1, k), (2, k), (3, l), (4, l)}

The function f is a many-one (as the two elements have the same image in Y) and it is onto (as every element of Y is the image of some element X). So, it is many-one onto function



FUNCTION (ALGEBRAICALLY)

ONE ONE FUNCTION

Let A and B are two sets , then f: A $\longrightarrow$ B is said to be one-one function if

$$f(x_1) = f(x_2)\ then\ x_1 = x_2$$
$$f(x_1) \neq f(x_2)\ then\ x_1 \neq x_2$$

Example

A	B	(A, B)
1	A	(1,A)
2	B	(2,B)
3	A	(3,A)
4	A	(4,B)

F (1) = F (3)

A = A

BUT 1 ≠ 3

It is not ON-ONE function

Example

A	B	(A, B)
1	1	(1,1)
2	3	(2,3)
3	2	(3,2)
4	4	(4,4)

F (1) ≠ F (3)

1 = 2

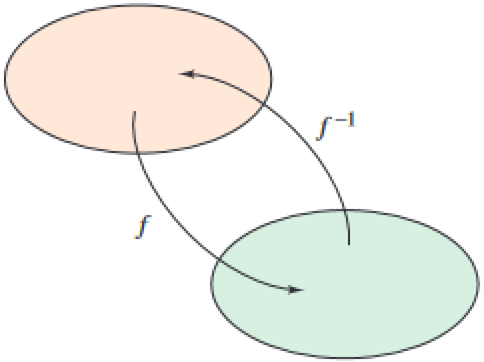
BUT 1 ≠ 2

It is ON-ONE function

EXAMPLES OF ONE-ONE FUNCTION

$f(x) = 3x - 1$ $f(x_1) = f(x_2)$ $3x_1 - 1 = 3x_2 - 1$ $3x_1 = 3x_2$ $x_1 = x_2$	$f(x) = 3x + 2$ $f(x_1) = f(x_2)$ $3x_1 + 2 = 3x_2 + 2$ $3x_1 = 3x_2$ $x_1 = x_2$
$f(x) = -3x^3 - 1$ $f(x_1) = f(x_2)$ $-3x_1^3 - 1 = -3x_2^3 - 1$ $-3x_1^3 = -3x_2^3$ $x_1 = x_2$	$f(x) = x^3 - x$ $f(x_1) = f(x_2)$ $x_1^3 - x = x_2^3 - x$ $x_1^3 - x_1^3 = x_1 - x_2$ $(x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2) = x_1 - x_2$ $(x_1 - x_2) - x_2$ $(x_1^2 + x_1x_2 + x_2^2) = x_1$ No real value

INVERSE FUNCTION



Domain of f = range of f^{-1}
Domain of f^{-1} = range of f
Figure 5.10

Inverse Functions

Recall from Section P.3 that a function can be represented by a set of ordered pairs. For instance, the function $f(x) = x + 3$ from $A = \{1, 2, 3, 4\}$ to $B = \{4, 5, 6, 7\}$ can be written as

$$f: \{(1, 4), (2, 5), (3, 6), (4, 7)\}.$$

By interchanging the first and second coordinates of each ordered pair, you can form the **inverse function** of f. This function is denoted by f^{-1}. It is a function from B to A, and can be written as

$$f^{-1}: \{(4, 1), (5, 2), (6, 3), (7, 4)\}.$$

Note that the domain of f is equal to the range of f^{-1}, and vice versa, as shown in Figure 5.10. The functions f and f^{-1} have the effect of "undoing" each other. That is, when you form the composition of f with f^{-1} or the composition of f^{-1} with f, you obtain the identity function.

$$f(f^{-1}(x)) = x \qquad \text{and} \qquad f^{-1}(f(x)) = x$$

Definition of Inverse Function

A function g is the **inverse function** of the function f if

$$f(g(x)) = x \quad \text{for each } x \text{ in the domain of } g$$

and

$$g(f(x)) = x \quad \text{for each } x \text{ in the domain of } f.$$

The function g is denoted by f^{-1} (read "f inverse").

Verifying Inverse Functions

Show that the functions are inverse functions of each other.

$$f(x) = 2x^3 - 1 \qquad \text{and} \qquad g(x) = \sqrt[3]{\frac{x+1}{2}}$$

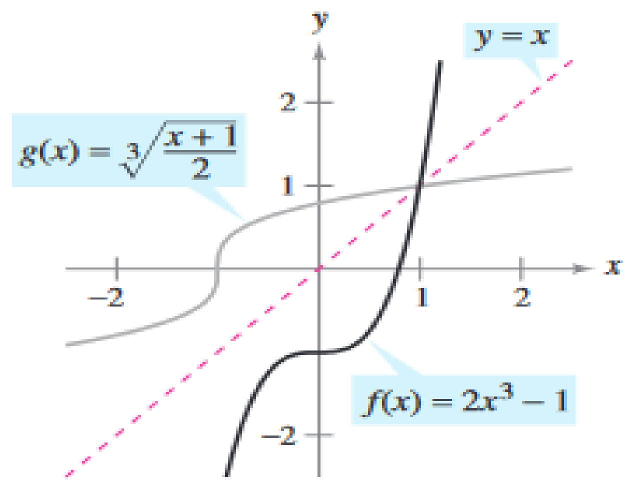
Solution Because the domains and ranges of both f and g consist of all real numbers, you can conclude that both composite functions exist for all x. The composition of f with g is given by

$$\begin{aligned} f(g(x)) &= 2\left(\sqrt[3]{\frac{x+1}{2}}\right)^3 - 1 \\ &= 2\left(\frac{x+1}{2}\right) - 1 \\ &= x + 1 - 1 \\ &= x. \end{aligned}$$

The composition of g with f is given by

$$\begin{aligned} g(f(x)) &= \sqrt[3]{\frac{(2x^3 - 1) + 1}{2}} \\ &= \sqrt[3]{\frac{2x^3}{2}} \\ &= \sqrt[3]{x^3} \\ &= x. \end{aligned}$$

Graphical way



f and g are inverse functions of each other.

Because $f(g(x)) = x$ and $g(f(x)) = x$, you can conclude that f and g are inverse

APPLICATION OF INVERSE FUNCTION

1. Solving Equations

One of the most common applications of inverse functions is solving equations where the variable is part of a more complex function. By using the inverse function, we can isolate the variable.

Example:

Suppose you have the equation $f(x) = y$, and you need to find x when given y.

- If $f(x) = 2x + 3$, the inverse function is $f^{-1}(x) = \frac{x-3}{2}$.

If you know $y = 7$, you can find x by applying the inverse function:

$$f^{-1}(7) = \frac{7-3}{2} = 2$$

Thus, $x = 2$.

APPLICATION OF INVERSE FUNCTION

2. Real-World Problems Involving Inverse Relationships

Inverse functions are useful in real-life scenarios where two quantities have a reverse relationship. For example:

- **Temperature Conversions:** Converting between Celsius and Fahrenheit is an example of inverse functions.
  - The formula for converting Fahrenheit to Celsius is:
  $$C = \frac{5}{9}(F - 32)$$
  - The inverse formula for converting Celsius to Fahrenheit is:
  $$F = \frac{9}{5}C + 32$$

  These formulas are inverse functions of each other.

Example:

If the temperature in Fahrenheit is $86^\circ F$, you can use the inverse function to find the temperature in Celsius:

$$C = \frac{5}{9}(86 - 32) = 30^\circ C$$

APPLICATION OF INVERSE FUNCTION

2. Solving Equations:

Inverse functions are used to solve equations when the original function is difficult to invert algebraically. If you have an equation $f(x) = y$, finding $f^{-1}(y)$ will give the value of x.

Example (Exponential and Logarithmic Functions):

For the exponential function $f(x) = e^x$, the inverse function is the natural logarithm $f^{-1}(x) = \ln(x)$.

- If you want to solve the equation $e^x = 5$, you apply the inverse function (logarithm) to both sides:

$$x = \ln(5)$$

Thus, inverse functions help solve equations involving exponential growth or decay, such as population growth, radioactive decay, or compound interest.

APPLICATION OF INVERSE FUNCTION

3. Coordinate Transformations in Geometry:

Inverse functions are used to **transform coordinates** between different systems. For example, converting Cartesian coordinates to polar coordinates involves an inverse function.

Example (Polar-Cartesian Coordinates Conversion):

The function to convert from polar coordinates (r, θ) to Cartesian coordinates (x, y) is:

$$x = r\cos(\theta), \quad y = r\sin(\theta)$$

The inverse function converts Cartesian coordinates back to polar:

$$r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

These conversions are commonly used in physics and engineering for simplifying the representation of problems in different coordinate systems.

APPLICATION OF INVERSE FUNCTION

4. Real-World Applications in Economics:

In economics, inverse functions help understand **supply and demand** relationships. If a function describes the demand for a product at a given price, the inverse function can provide the price at a certain demand level.

Example (Supply and Demand):

If the demand function is $D(p) = 100 - 5p$, where p is the price of a product, the inverse function $D^{-1}(q)$ gives the price p that corresponds to a certain quantity q of demand:

$$p = \frac{100 - q}{5}$$

This allows businesses to adjust pricing strategies based on desired demand.

APPLICATION OF INVERSE FUNCTION

5. Cryptography and Security (Public-Key Encryption):

Inverse functions are fundamental in cryptography, particularly in **public-key encryption** methods. One example is the RSA encryption algorithm, which is widely used for secure data transmission.

Example (RSA Encryption):

- RSA encryption uses two keys: a **public key** for encryption and a **private key** for decryption. The encryption function $E(x)$ uses the public key to encode the message, while the decryption function $D(x)$, which is the inverse of $E(x)$, uses the private key to recover the original message.

If a message is encrypted using $E(x)$, the inverse function $D(x)$ is applied to get the original message back.

APPLICATION OF INVERSE FUNCTION

6. Physics (Kinematics and Motion):

In physics, inverse functions are useful for reversing processes like velocity and position relationships.

Example (Position-Velocity Relationship):

If the position of an object $s(t)$ at time t is given by a function, the inverse function can be used to find the time when the object reaches a certain position.

For example, if $s(t) = 5t^2$, then the inverse function tells you at what time t the object will be at a given position s. In this case:

$$t = \sqrt{\frac{s}{5}}$$

7. Electrical Engineering (Signal Processing):

Inverse functions are used in signal processing to **recover original signals** after they have been transformed or encoded.

Example (Fourier Transform):

The Fourier transform converts a time-domain signal into its frequency components. The **inverse Fourier transform** is used to recover the original time-domain signal from the frequency-domain representation.

15. Determine whether the function $f: \mathbf{Z} \times \mathbf{Z} \to \mathbf{Z}$ is onto if

a) $f(m, n) = m + n$.
b) $f(m, n) = m^2 + n^2$.
c) $f(m, n) = m$.
d) $f(m, n) = |n|$.
e) $f(m, n) = m - n$.

15. An onto function is one whose range is the entire codomain. Thus we must determine whether we can write every integer in the form given by the rule for f in each case.

a) Given any integer n, we have $f(0, n) = n$, so the function is onto.

b) Clearly the range contains no negative integers, so the function is not onto.

c) Given any integer m, we have $f(m, 25) = m$, so the function is onto. (We could have used any constant in place of 25 in this argument.)

d) Clearly the range contains no negative integers, so the function is not onto.

e) Given any integer m, we have $f(m, 0) = m$, so the function is onto.

23. Determine whether each of these functions is a bijection from **R** to **R**.

a) $f(x) = 2x + 1$

b) $f(x) = x^2 + 1$

c) $f(x) = x^3$

d) $f(x) = (x^2 + 1)/(x^2 + 2)$

23. a) One way to determine whether a function is a bijection is to try to construct its inverse. This function is a bijection, since its inverse (obtained by solving $y = 2x + 1$ for x) is the function $g(y) = (y - 1)/2$. Alternatively, we can argue directly. To show that the function is one-to-one, note that if $2x + 1 = 2x' + 1$, then $x = x'$. To show that the function is onto, note that $2((y - 1)/2) + 1 = y$, so every number is in the range.

b) This function is not a bijection, since its range is the set of real numbers greater than or equal to 1 (which is sometimes written $[1, \infty)$), not all of $\mathbf{R}$. (It is not injective either.)

c) This function is a bijection, since it has an inverse function, namely the function $f(y) = y^{1/3}$ (obtained by solving $y = x^3$ for x).

d) This function is not a bijection. It is easy to see that it is not injective, since x and $-x$ have the same image, for all real numbers x. A little work shows that the range is only $\{ y \mid 0.5 \leq y < 1 \} = [0.5, 1)$.

31. Let $f(x) = \lfloor x^2/3 \rfloor$. Find $f(S)$ if

a) $S = \{-2, -1, 0, 1, 2, 3\}$.

b) $S = \{0, 1, 2, 3, 4, 5\}$.

c) $S = \{1, 5, 7, 11\}$.

d) $S = \{2, 6, 10, 14\}$.

31. In each case, we need to compute the values of $f(x)$ for each $x \in S$.

a) Note that $f(\pm 2) = \lfloor (\pm 2)^2/3 \rfloor = \lfloor 4/3 \rfloor = 1$, $f(\pm 1) = \lfloor (\pm 1)^2/3 \rfloor = \lfloor 1/3 \rfloor = 0$, $f(0) = \lfloor 0^2/3 \rfloor = \lfloor 0 \rfloor = 0$, and $f(3) = \lfloor 3^2/3 \rfloor = \lfloor 3 \rfloor = 3$. Therefore $f(S) = \{0, 1, 3\}$.

b) In addition to the values we computed above, we note that $f(4) = 5$ and $f(5) = 8$. Therefore $f(S) = \{0, 1, 3, 5, 8\}$.

c) Note this time also that $f(7) = 16$ and $f(11) = 40$, so $f(S) = \{0, 8, 16, 40\}$.

d) $\{f(2), f(6), f(10), f(14)\} = \{1, 12, 33, 65\}$

77. For each of these partial functions, determine its domain, codomain, domain of definition, and the set of values for which it is undefined. Also, determine whether it is a total function.

a) $f : \mathbf{Z} \to \mathbf{R}$, $f(n) = 1/n$
b) $f : \mathbf{Z} \to \mathbf{Z}$, $f(n) = \lceil n/2 \rceil$
c) $f : \mathbf{Z} \times \mathbf{Z} \to \mathbf{Q}$, $f(m, n) = m/n$
d) $f : \mathbf{Z} \times \mathbf{Z} \to \mathbf{Z}$, $f(m, n) = mn$
e) $f : \mathbf{Z} \times \mathbf{Z} \to \mathbf{Z}$, $f(m, n) = m - n$ if $m > n$

77. In each case we easily read the domain and codomain from the notation. The domain of definition is obtained by determining for which values the definition makes sense. The function is total if the domain of definition is the entire domain (so that there are no values for which the partial function is undefined).

a) The domain is $\mathbf{Z}$ and the codomain is $\mathbf{R}$. Since division is possible by every nonzero number, the domain of definition is all the nonzero integers; $\{0\}$ is the set of values for which f is undefined. (It is not total.)

b) The domain and codomain are both given to be $\mathbf{Z}$. Since the definition makes sense for all integers, this is a total function, whose domain of definition is also $\mathbf{Z}$; the set of values for which f is undefined is $\emptyset$.

c) By inspection, the domain is the Cartesian product $\mathbf{Z} \times \mathbf{Z}$, and the codomain is $\mathbf{Q}$. Since fractions cannot have a 0 in the denominator, we must exclude the "slice" of $\mathbf{Z} \times \mathbf{Z}$ in which the second coordinate is 0. Thus the domain of definition is $\mathbf{Z} \times (\mathbf{Z} - \{0\})$, and the function is undefined for all values in $\mathbf{Z} \times \{0\}$. It is not a total function.

d) The domain is given to be $\mathbf{Z} \times \mathbf{Z}$ and the codomain is given to be $\mathbf{Z}$. Since the definition makes sense for all pairs of integers, this is a total function, whose domain of definition is also $\mathbf{Z} \times \mathbf{Z}$; the set of values for which f is undefined is $\emptyset$.

e) Again the domain and codomain are $\mathbf{Z} \times \mathbf{Z}$ and $\mathbf{Z}$, respectively. Since the definition is only stated for those pairs in which the first coordinate exceeds the second, the domain of definition is $\{ (m, n) \mid m > n \}$, and therefore the set of values for which the function is undefined is $\{ (m, n) \mid m \leq n \}$. It is not a total function.

Recursive Relations and Examples

A recurrence relation (recursive relation) is an equation that expresses a term in a sequence as a function of its preceding terms. It is commonly used in mathematics, algorithms, and computer science to describe problems that can be broken down into smaller subproblems.

General Form of a Recurrence Relation

A recurrence relation expresses $T(n)$ in terms of smaller values:

$$T(n) = f(T(n-1), T(n-2), \ldots, T(1))$$

where f is a function that describes the recurrence.

1. Fibonacci Sequence

The Fibonacci sequence is defined as:

$$F(n) = F(n-1) + F(n-2)$$

with base cases:

$$F(0) = 0, \quad F(1) = 1$$

Example Calculation

$$F(5) = F(4) + F(3)$$

$$F(4) = F(3) + F(2)$$

$$F(3) = F(2) + F(1)$$

2. Factorial Function

The factorial of a number n is given by:

$$n! = n \times (n-1)!$$

with base case:

$$0! = 1$$

Example Calculation

$$5! = 5 \times 4!$$

$$4! = 4 \times 3!$$

$$3! = 3 \times 2!$$

3. Tower of Hanoi (Move Count)

The Tower of Hanoi puzzle requires moving n disks from one peg to another using a third peg, following the rule that only one disk can be moved at a time, and no larger disk can be placed on a smaller one.

The recurrence relation for the **minimum number of moves** is:

$$T(n) = 2T(n-1) + 1$$

with base case:

$$T(1) = 1$$

Example Calculation

$$T(2) = 2T(1) + 1 = 3$$

4. Binary Search (Time Complexity)

Binary Search is a divide-and-conquer algorithm where the search space is divided in half at each step.

The recurrence relation is:

$$T(n) = T(n/2) + O(1)$$

with base case:

$$T(1) = O(1)$$

Example Calculation

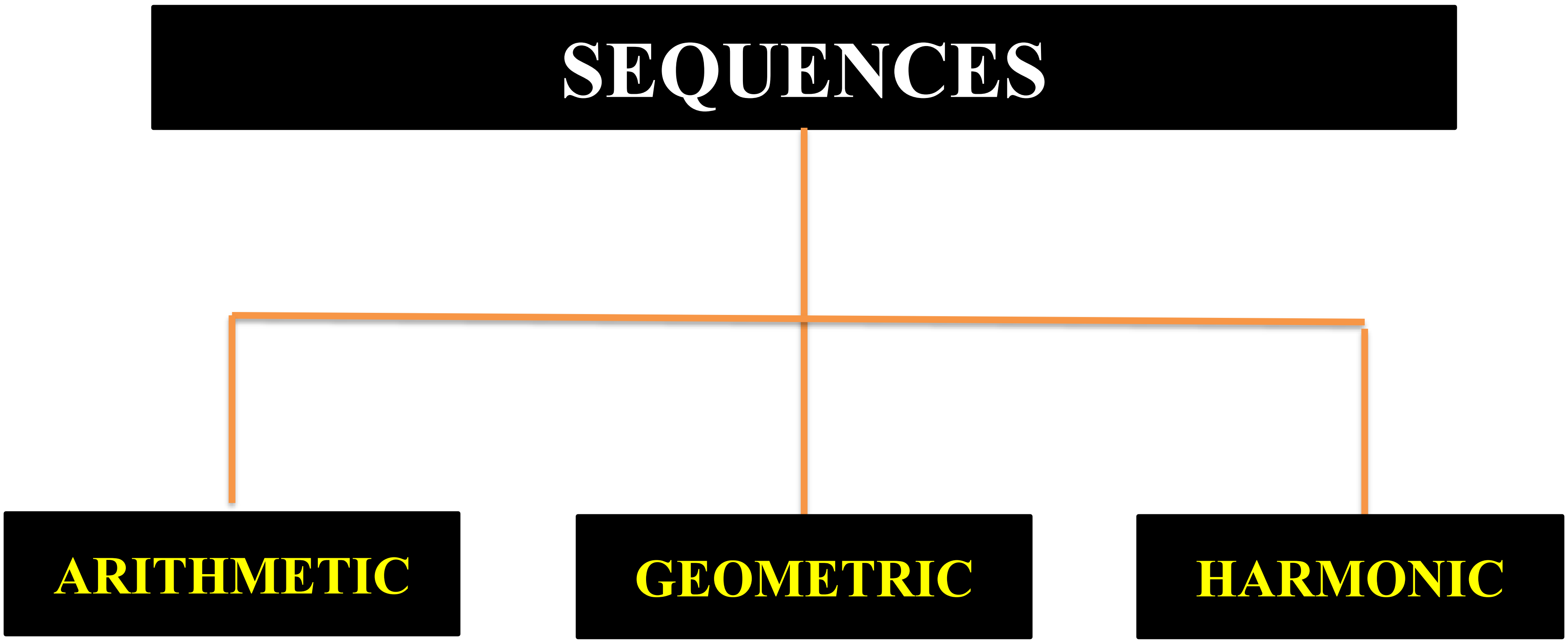
$$T(8) = T(4) + O(1)$$

$$T(4) = T(2) + O(1)$$

$$T(2) = T(1) + O(1)$$

Using **recursion tree analysis**, we get **T(n) = O(\log n)**.

SEQUENCES AND SUMMATIONS


SEQUENCES
ARITHMETIC
GEOMETRIC
HARMONIC

SEQUENCE:

A sequence is just a list of elements usually written in a row.

EXAMPLES:

1. 1, 2, 3, 4, 5, …
2. 4, 8, 12, 16, 20,…
3. 2, 4, 8, 16, 32, …
4. 1, 1/2, 1/3, 1/4, 1/5, …
5. 1, 4, 9, 16, 25, …
6. 1, -1, 1, -1, 1, -1, …

NOTE: The symbol "…" is called ellipsis, and reads "and so forth"

FORMAL DEFINITION:

A sequence is a function whose domain is the set of integers greater than or equal to a particular integer n_0. Usually this set is the set of Natural numbers {1, 2, 3, …} or the set of whole numbers {0, 1, 2, 3, …}.

NOTATION:

We use the notation a_n to denote the image of the integer n, and call it a term of the sequence. Thus

$$a_1, a_2, a_3, a_4 \dots, a_n, \dots$$

represent the terms of a sequence defined on the set of natural numbers N.
Note that a sequence is described by listing the terms of the sequence in order of increasing subscripts.

EXAMPLE:

Write the first four terms of the sequence defined by the formula

$$b_j = 1 + 2^j, \text{ for all integers } j \geq 0$$

SOLUTION:

$b_0 = 1 + 2^0 = 1 + 1 = 2$
$b_1 = 1 + 2^1 = 1 + 2 = 3$
$b_2 = 1 + 2^2 = 1 + 4 = 5$
$b_3 = 1 + 2^3 = 1 + 8 = 9$

REMARK:

The formula $b_j = 1 + 2^j$, for all integers $j \geq 0$ defines an infinite sequence having infinite number of values.

EXERCISE:

Compute the first six terms of the sequence defined by the formula

$$C_n = 1 + (-1)^n \text{ for all integers } n \geq 0$$

SOLUTION :

$C_0 = 1 + (-1)^0 = 1 + 1 = 2$ $\qquad$ $C_1 = 1 + (-1)^1 = 1 + (-1) = 0$
$C_2 = 1 + (-1)^2 = 1 + 1 = 2$ $\qquad$ $C_3 = 1 + (-1)^3 = 1 + (-1) = 0$
$C_4 = 1 + (-1)^4 = 1 + 1 = 2$ $\qquad$ $C_5 = 1 + (-1)^5 = 1 + (-1) = 0$

REMARK:

1) If n is even, then $C_n = 2$ and if n is odd, then $C_n = 0$
   Hence, the sequence oscillates endlessly between 2 and 0.
2) An infinite sequence may have only a finite number of values.

ARITHMETIC SEQUENCE:

A sequence in which every term after the first is obtained from the preceding term by adding a constant number is called an arithmetic sequence or arithmetic progression (A.P.)

The constant number, being the difference of any two consecutive terms is called the common difference of A.P., commonly denoted by "d".

EXAMPLES:

1. $5, 9, 13, 17, \ldots$ (common difference = 4)
2. $0, -5, -10, -15, \ldots$ (common difference = -5)
3. $x + a, x + 3a, x + 5a, \ldots$ (common difference = 2a)

GENERAL TERM OF AN ARITHMETIC SEQUENCE:

Let **a** be the first term and **d** be the common difference of an arithmetic sequence. Then the sequence is $a, a+d, a+2d, a+3d, \ldots$

If a_i, for $i \geq 1$, represents the terms of the sequence then

$$a_1 = \text{first term} = a = a + (1-1)\,d$$

$$a_2 = \text{second term} = a + d = a + (2-1)\,d$$

$$a_3 = \text{third term} = a + 2d = a + (3-1)\,d$$

By symmetry

$$a_n = n\text{th term} = a + (n-1)d \text{ for all integers } n \geq 1.$$

EXAMPLE:

Find the 20th term of the arithmetic sequence

3, 9, 15, 21, …

SOLUTION:

Here a = first term = 3

d = common difference = 9 - 3 = 6

n = term number = 20

a_{20} = value of 20th term = ?

Since $a_n = a + (n - 1) d; \quad n \geq 1$

$\therefore \quad a_{20} = 3 + (20 - 1) 6$

$= 3 + 114$

$= 117$

EXAMPLE:

Which term of the arithmetic sequence

4, 1, -2, …, is -77

SOLUTION:

Here a = first term = 4

d = common difference = 1 - 4 = -3

a_n = value of nth term = - 77

n = term number = ?

Since

$an = a + (n - 1) d \quad n \geq 1$

$\Rightarrow \quad -77 = 4 + (n - 1)(-3)$

$\Rightarrow \quad -77 - 4 = (n - 1)(-3)$

OR

$$\frac{-81}{-3} = n - 1$$

OR

$27 = n - 1$

$n = 28$

Hence –77 is the 28th term of the given sequence.

EXERCISE:

Find the 36th term of the arithmetic sequence whose 3rd term is 7 and 8th term is 17.

SOLUTION:

Let **a** be the first term and **d** be the common difference of the arithmetic sequence.

Then

$a_n = a + (n - 1)d \qquad n \geq 1$

$\Rightarrow \quad a_3 = a + (3 - 1)\, d$

and $\quad a_8 = a + (8 - 1)\, d$

Given that $a_3 = 7$ and $a_8 = 17$. Therefore

$7 = a + 2d$……………………(1)

and $\quad 17 = a + 7d$……………………(2)

Subtracting (1) from (2), we get,

$10 = 5d$

$\Rightarrow \quad d = 2$

Substituting $d = 2$ in (1) we have

$7 = a + 2(2)$

which gives $a = 3$

Thus, $\quad a_n = a + (n - 1)\, d$

$a_n = 3 + (n - 1)\, 2 \qquad$ (using values of a and d)

Hence the value of 36th term is

$$\begin{aligned} a_{36} &= 3 + (36 - 1)\, 2 \\ &= 3 + 70 \\ &= 73 \end{aligned}$$

GEOMETRIC SEQUENCE:

A sequence in which every term after the first is obtained from the preceding term by multiplying it with a constant number is called a geometric sequence or geometric progression (G.P.)

The constant number, being the ratio of any two consecutive terms is called the common ratio of the G.P. commonly denoted by "r".

EXAMPLE:

1. $1, 2, 4, 8, 16, \ldots$ (common ratio = 2)
2. $3, -3/2, 3/4, -3/8, \ldots$ (common ratio = - 1/2)
3. $0.1, 0.01, 0.001, 0.0001, \ldots$ (common ratio = 0.1 = 1/10)

GENERAL TERM OF A GEOMETRIC SEQUENCE:

Let **a** be the first tem and **r** be the common ratio of a geometric sequence. Then the sequence is $a, ar, ar^2, ar^3, \ldots$

If a_i, for $i \geq 1$ represent the terms of the sequence, then

a_1 = first term = $a = ar^{1-1}$

a_2 = second term = $ar = ar^{2-1}$

a_3 = third term = $ar^2 = ar^{3-1}$

....

....

a_n = nth term = ar^{n-1}; for all integers $n \geq 1$

EXERCISE:

Write the geometric sequence with positive terms whose second term is 9 and fourth term is 1.

SOLUTION:

Let **a** be the first term and **r** be the common ratio of the geometric sequence. Then

$a_n = ar^{n-1}$ $\qquad n \geq 1$

Now $\qquad a_2 = ar^{2-1}$

$\Rightarrow \qquad 9 = ar$ ....(1)

Also $\qquad a_4 = ar^{4-1}$

$\qquad 1 = ar^3$ ....(2)

Dividing (2) by (1), we get,

$$\frac{1}{9} = \frac{ar^3}{ar}$$

$$\Rightarrow \quad \frac{1}{9} = r^2$$

$$\Rightarrow \quad r = \frac{1}{3} \qquad \left(\text{rejecting } r = -\frac{1}{3}\right)$$

Substituting r = 1/3 in (1), we get

$$9 = a\left(\frac{1}{3}\right)$$

$$\Rightarrow \quad a = 9 \times 3 = 27$$

Hence the geometric sequence is

27, 9, 3, 1, 1/3, 1/9, …

INFINITE GEOMETRIC SERIES:

Consider the infinite geometric series

$$a + ar + ar^2 + \ldots + ar^{n-1} + \ldots$$

then

$$S_n = a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a(1-r^n)}{1-r} \qquad (r \neq 1)$$

If $S_n \to S$ as $n \to \infty$, then the series is convergent and S is its sum.

If $|r| < 1$, then $r^n \to 0$ as $n \to \infty$

$$\therefore \qquad S = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r}$$

$$= \frac{a}{1-r}$$

If S_n increases indefinitely as n becomes very large then the series is said to be divergent.

EXERCISE:

Find the sum of the infinite geometric series:

$$\frac{9}{4}+\frac{3}{2}+1+\frac{2}{3}+\cdots$$

SOLUTION:

Here we have

$$a=\frac{9}{4}, \quad r=\frac{3/2}{9/4}=\frac{2}{3}$$

Note that |r| < 1 So we can use the above formula.

$$\begin{aligned}
\therefore \qquad S &= \frac{a}{1-r} \\
&= \frac{9/4}{1-2/3} \\
&= \frac{9/4}{1/3}=\frac{9}{4}\times\frac{3}{1}=\frac{27}{4}
\end{aligned}$$

EXERCISE:

Find a common fraction for the recurring decimal 0.81

SOLUTION:

$$0.81 = 0.8181818181 \ldots$$
$$= 0.81 + 0.0081 + 0.000081 + \ldots$$

which is an infinite geometric series with

$$a = 0.81, \quad r = \frac{0.0081}{0.81} = 0.01$$

$$\therefore \quad \text{Sum} = \frac{a}{1-r}$$
$$= \frac{0.81}{1-0.01} = \frac{0.81}{0.99}$$
$$= \frac{81}{99} = \frac{9}{11}$$

Exercises

1. Find these terms of the sequence $\{a_n\}$, where $a_n = 2 \cdot (-3)^n + 5^n$.

a) a_0 **b)** a_1 **c)** a_4 **d)** a_5

2. What is the term a_8 of the sequence $\{a_n\}$ if a_n equals

a) 2^{n-1}? **b)** 7?
c) $1 + (-1)^n$? **d)** $-(-2)^n$?

3. What are the terms a_0, a_1, a_2, and a_3 of the sequence $\{a_n\}$, where a_n equals

a) $2^n + 1$? **b)** $(n+1)^{n+1}$?
c) $\lfloor n/2 \rfloor$? **d)** $\lfloor n/2 \rfloor + \lceil n/2 \rceil$?

4. What are the terms a_0, a_1, a_2, and a_3 of the sequence $\{a_n\}$, where a_n equals

a) $(-2)^n$? **b)** 3?
c) $7 + 4^n$? **d)** $2^n + (-2)^n$?

5. List the first 10 terms of each of these sequences.

a) the sequence that begins with 2 and in which each successive term is 3 more than the preceding term

b) the sequence that lists each positive integer three times, in increasing order

c) the sequence that lists the odd positive integers in increasing order, listing each odd integer twice

d) the sequence whose nth term is $n! - 2^n$

e) the sequence that begins with 3, where each succeeding term is twice the preceding term

f) the sequence whose first term is 2, second term is 4, and each succeeding term is the sum of the two preceding terms

g) the sequence whose nth term is the number of bits in the binary expansion of the number n (defined in Section 4.2)

h) the sequence where the nth term is the number of letters in the English word for the index n

6. List the first 10 terms of each of these sequences.

a) the sequence obtained by starting with 10 and obtaining each term by subtracting 3 from the previous term

b) the sequence whose nth term is the sum of the first n positive integers

c) the sequence whose nth term is $3^n - 2^n$

d) the sequence whose nth term is $\lfloor \sqrt{n} \rfloor$

e) the sequence whose first two terms are 1 and 5 and each succeeding term is the sum of the two previous terms

f) the sequence whose nth term is the largest integer whose binary expansion (defined in Section 4.2) has n bits (Write your answer in decimal notation.)

g) the sequence whose terms are constructed sequentially as follows: start with 1, then add 1, then multiply by 1, then add 2, then multiply by 2, and so on

h) the sequence whose nth term is the largest integer k such that $k! \leq n$

7. Find at least three different sequences beginning with the terms 1, 2, 4 whose terms are generated by a simple formula or rule.

8. Find at least three different sequences beginning with the terms 3, 5, 7 whose terms are generated by a simple formula or rule.

9. Find the first five terms of the sequence defined by each of these recurrence relations and initial conditions.

a) $a_n = 6a_{n-1}$, $a_0 = 2$

b) $a_n = a_{n-1}^2$, $a_1 = 2$

c) $a_n = a_{n-1} + 3a_{n-2}$, $a_0 = 1$, $a_1 = 2$

d) $a_n = na_{n-1} + n^2a_{n-2}$, $a_0 = 1$, $a_1 = 1$

e) $a_n = a_{n-1} + a_{n-3}$, $a_0 = 1$, $a_1 = 2$, $a_2 = 0$

0. Find the first six terms of the sequence defined by each of these recurrence relations and initial conditions.

a) $a_n = -2a_{n-1}$, $a_0 = -1$

b) $a_n = a_{n-1} - a_{n-2}$, $a_0 = 2$, $a_1 = -1$

c) $a_n = 3a_{n-1}^2$, $a_0 = 1$

d) $a_n = na_{n-1} + a_{n-2}^2$, $a_0 = -1$, $a_1 = 0$

e) $a_n = a_{n-1} - a_{n-2} + a_{n-3}$, $a_0 = 1$, $a_1 = 1$, $a_2 = 2$

1. Let $a_n = 2^n + 5 \cdot 3^n$ for $n = 0, 1, 2, \ldots$.

a) Find a_0, a_1, a_2, a_3, and a_4.

b) Show that $a_2 = 5a_1 - 6a_0$, $a_3 = 5a_2 - 6a_1$, and $a_4 = 5a_3 - 6a_2$.

c) Show that $a_n = 5a_{n-1} - 6a_{n-2}$ for all integers n with $n \geq 2$.

2. Show that the sequence $\{a_n\}$ is a solution of the recurrence relation $a_n = -3a_{n-1} + 4a_{n-2}$ if

a) $a_n = 0$. **b)** $a_n = 1$.

c) $a_n = (-4)^n$. **d)** $a_n = 2(-4)^n + 3$.

3. Is the sequence $\{a_n\}$ a solution of the recurrence relation $a_n = 8a_{n-1} - 16a_{n-2}$ if

a) $a_n = 0$? **b)** $a_n = 1$?

c) $a_n = 2^n$? **d)** $a_n = 4^n$?

e) $a_n = n4^n$? **f)** $a_n = 2 \cdot 4^n + 3n4^n$?

g) $a_n = (-4)^n$? **h)** $a_n = n^2 4^n$?

c) $a_n = 3(-1)^n + 2^n - n + 2$.

d) $a_n = 7 \cdot 2^n - n + 2$.

16. Find the solution to each of these recurrence relations with the given initial conditions. Use an iterative approach such as that used in Example 10.

a) $a_n = -a_{n-1}$, $a_0 = 5$

b) $a_n = a_{n-1} + 3$, $a_0 = 1$

c) $a_n = a_{n-1} - n$, $a_0 = 4$

d) $a_n = 2a_{n-1} - 3$, $a_0 = -1$

e) $a_n = (n + 1)a_{n-1}$, $a_0 = 2$

f) $a_n = 2na_{n-1}$, $a_0 = 3$

g) $a_n = -a_{n-1} + n - 1$, $a_0 = 7$

17. Find the solution to each of these recurrence relations and initial conditions. Use an iterative approach such as that used in Example 10.

a) $a_n = 3a_{n-1}$, $a_0 = 2$

b) $a_n = a_{n-1} + 2$, $a_0 = 3$

c) $a_n = a_{n-1} + n$, $a_0 = 1$

d) $a_n = a_{n-1} + 2n + 3$, $a_0 = 4$

e) $a_n = 2a_{n-1} - 1$, $a_0 = 1$

f) $a_n = 3a_{n-1} + 1$, $a_0 = 1$

g) $a_n = na_{n-1}$, $a_0 = 5$

h) $a_n = 2na_{n-1}$, $a_0 = 1$

18. A person deposits \$1000 in an account that yields 9% interest compounded annually.

a) Set up a recurrence relation for the amount in the account at the end of n years.

b) Find an explicit formula for the amount in the account at the end of n years.

c) How much money will the account contain after 100 years?

19. Suppose that the number of bacteria in a colony triples every hour.

a) Set up a recurrence relation for the number of bacteria after n hours have elapsed.

b) If 100 bacteria are used to begin a new colony, how many bacteria will be in the colony in 10 hours?

20. Assume that the population of the world in 2010 was 6.9 billion and is growing at the rate of 1.1% a year.

a) Set up a recurrence relation for the population of the world n years after 2010.

b) Find an explicit formula for the population of the world n years after 2010.

c) What will the population of the world be in 2030?

SUMMATION

SUMMATION NOTATION:

The capital Greek letter sigma $\sum$ is used to write a sum in a short hand notation.
where k varies from 1 to n represents the sum given in expanded form by
$= a_1 + a_2 + a_3 + \ldots + a_n$
More generally if m and n are integers and $m \le n$, then the summation from k equal m to n of a_k is

$$\sum_{k=m}^{n} a_k = a_m + a_{m+1} + a_{m+2} + \cdots + a_n$$

Here **k** is called the index of the summation; **m** the lower limit of the summation and **n** the upper limit of the summation.

COMPUTING SUMMATIONS:

Let $a_0 = 2$, $a_1 = 3$, $a_2 = -2$, $a_3 = 1$ and $a_4 = 0$.Compute each of the summations:

$$(a) \quad \sum_{i=0}^{4} a_i \qquad (b) \quad \sum_{j=0}^{2} a_{2j} \qquad (c) \quad \sum_{k=1}^{1} a_k$$

SOLUTION:

(a) $\displaystyle\sum_{i=0}^{4} a_i = a_0 + a_1 + a_2 + a_3 + a_4$
$= 2 + 3 + (-2) + 1 + 0 = 4$

(b) $\displaystyle\sum_{j=0}^{2} a_{2j} = a_0 + a_2 + a_4$
$= 2 + (-2) + 0 = 0$

(c) $\displaystyle\sum_{k=1}^{1} a_k = a_1$
$= 3$

EXERCISE:

Compute the summations

1. $$\sum_{i=1}^{3}(2i-1) = [2(1)-1]+[2(2)-1]+[2(3)-1]$$

$$= 1+3+5$$

$$= 9$$

2. $$\sum_{k=-1}^{1}(k^3+2) = [(-1)^3+2]+[(0)^3+2]+[(1)^3+2]$$

$$= [-1+2]+[0+2]+[1+2]$$

$$= 1+2+3$$

$$= 6$$

SUMMATION NOTATION TO EXPANDED FORM:

Write the summation $\sum_{i=0}^{n} \frac{(-1)^i}{i+1}$ to expanded form.

SOLUTION:

$$\sum_{i=0}^{n} \frac{(-1)^i}{i+1} = \frac{(-1)^0}{0+1} + \frac{(-1)^1}{1+1} + \frac{(-1)^2}{2+1} + \frac{(-1)^3}{3+1} + \cdots + \frac{(-1)^n}{n+1}$$

$$= \frac{1}{1} + \frac{(-1)}{2} + \frac{1}{3} + \frac{(-1)}{4} + \cdots + \frac{(-1)^n}{n+1}$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{(-1)^n}{n+1}$$

PROPERTIES OF SUMMATIONS:

1. $\sum_{k=m}^{n}(a_k+b_k) = \sum_{k=m}^{n}a_k+\sum_{k=m}^{n}b_k; \quad a_k, b_k \in R$

2. $\sum_{k=m}^{n}ca_k = c\sum_{k=m}^{n}a_k \quad$ c∈R

3. $\sum_{k=a-i}^{b-i}(k+i) = \sum_{k=a}^{b}k \quad i \in N$

4. $\sum_{k=a+i}^{b+i}(k-i) = \sum_{k=a}^{b}k \quad i \in N$

5. $\sum_{k=1}^{n}c = c+c+\cdots+c = nc$

EXERCISE:

Express the following summation more simply:

SOLUTION: $3\sum_{k=1}^{n}(2k-3)+\sum_{k=1}^{n}(4-5k)$

$$3\sum_{k=1}^{n}(2k-3)+\sum_{k=1}^{n}(4-5k)$$

$$=3\sum_{k=1}^{n}3(2k-3)+\sum_{k=1}^{n}(4-5k)$$

$$=\sum_{k=1}^{n}[3(2k-3)+(4-5k)]$$

$$=\sum_{k=1}^{n}(k-5)$$

$$=\sum_{k=1}^{n}k-\sum_{k=1}^{n}5$$

$$=\sum_{k=1}^{n}k-5n$$

30. What are the values of these sums, where $S = \{1, 3, 5, 7\}$?

a) $\sum_{j \in S} j$ **b)** $\sum_{j \in S} j^2$

c) $\sum_{j \in S} (1/j)$ **d)** $\sum_{j \in S} 1$

31. What is the value of each of these sums of terms of a geometric progression?

a) $\sum_{j=0}^{8} 3 \cdot 2^j$ **b)** $\sum_{j=1}^{8} 2^j$

c) $\sum_{j=2}^{8} (-3)^j$ **d)** $\sum_{j=0}^{8} 2 \cdot (-3)^j$

32. Find the value of each of these sums.

a) $\sum_{j=0}^{8} (1 + (-1)^j)$ **b)** $\sum_{j=0}^{8} (3^j - 2^j)$

c) $\sum_{j=0}^{8} (2 \cdot 3^j + 3 \cdot 2^j)$ **d)** $\sum_{j=0}^{8} (2^{j+1} - 2^j)$

33. Compute each of these double sums.

a) $\sum_{i=1}^{2} \sum_{j=1}^{3} (i + j)$ **b)** $\sum_{i=0}^{2} \sum_{j=0}^{3} (2i + 3j)$

c) $\sum_{i=1}^{3} \sum_{j=0}^{2} i$ **d)** $\sum_{i=0}^{2} \sum_{j=1}^{3} ij$

34. Compute each of these double sums.

a) $\sum_{i=1}^{3} \sum_{j=1}^{2} (i - j)$ **b)** $\sum_{i=0}^{3} \sum_{j=0}^{2} (3i + 2j)$

c) $\sum_{i=1}^{3} \sum_{j=0}^{2} j$ **d)** $\sum_{i=0}^{2} \sum_{j=0}^{3} i^2 j^3$